CWM Macro-Scale Experiment Guide

Mike Tierce

Independent Researcher

ORCID: 0009-0004-3869-958X

github.com/miketierce/wcfoma

Version

About This Document

This companion document contains the complete macro-scale experiment guide, full-scale illustration plates, and printable data worksheets for reproducing the Coherent Wave Memory (CWM) prototype experiments described in Section 4 of the main paper.

The guide is self-contained: every component is listed with a direct purchase link, every procedure is numbered for reproducibility, and every known failure mode includes a tested mitigation. A middle school science teacher with no acoustics background should be able to build the prototype, complete all eight experiments, and contribute publishable data within a single school week. See Section 4 of the main paper for the theoretical context behind each measurement.

All quantitative claims in the main paper are computed from first-principles simulation code (34 modules, 1,036 automated tests, all passing) and independently validated by finite element analysis. No curve fitting, no adjusted parameters, no post-hoc corrections. Repository: github.com/miketierce/wcfoma.

Appendix D: Macro-Scale Experiment Guide

This appendix provides step-by-step instructions for replicating the macro-scale prototype experiments of Section 4. The guide is designed to be self-contained: every component is listed with a direct purchase link, every procedure is numbered for reproducibility, and every known failure mode includes a tested mitigation. A middle school science teacher with no acoustics background should be able to build the prototype, complete all eight experiments, and contribute publishable data within a single school week. See Section 4 for the theoretical context behind each measurement.

The glass harmonica connection. Every experiment in this appendix is a direct descendant of a musical instrument that predates the transistor by centuries. A glass harmonica—tuned wineglasses played by rubbing a wet finger around the rim—demonstrates every principle of CWM in audible form: glass resonators with eigenfrequencies set by geometry, mass perturbation tuning via water level, continuous-wave excitation via stick-slip friction, and spectral readout by the human ear. In 1761, Benjamin Franklin attended a glass harmonica concert in London and built an improved version: the *glass armonica*, which mounted the bowls on a rotating spindle so a performer could vary finger pressure, position, and contact duration in real time. Same glass, same physics, same resonant modes—but now reconfigurable. This appendix walks the same path: Experiments 1–6 build and characterize a fixed resonator (the harmonica); Experiment 7 demonstrates continuous-wave precision readout (bowing vs. ringing); and Experiment 8 demonstrates rewritable encoding with water drops (the armonica).

D.1 Complete Bill of Materials

The core BOM from Section 4.2 is expanded below with recommended quantities (extras for breakage and controls), supplier links, and additional items needed for failure-mode mitigations. All prices are approximate as of 2026 and may vary.

Table D.1: Core Components

#	Component	Specification	Qty	Est. Cost	Amazon Link
1	Borosilicate glass stirring rods	6 mm dia × 150 mm (5.9″), rounded ends, Boro 3.3	15-pack	~$8	PATIKIL 15 Pcs, 5.9″ × 6 mm
2	Borosilicate glass stirring rods (alt.)	6 mm dia × 200 mm (7.9″), Boro 3.3—cut to 150 mm if needed	10-pack	~$9	EISCO 10PK, 7.9″ × 6 mm
3	Piezoelectric discs with leads	10 mm dia, PZT ceramic, pre-soldered 4″ wire leads	15-pack	~$7	E-outstanding 15 PCS, 10 mm
4	Piezoelectric discs (alt.)	10 mm dia, PZT, bare discs (solder leads yourself)	10-pack	~$6	uxcell 10 Pcs, 10 mm
5	USB oscilloscope + waveform generator	PicoScope 2204A, 10 MHz BW, 2-ch, built-in AWG, PS7 software	1	~$192	PicoScope 2204A
6	Cyanoacrylate glue (super glue)	Medium viscosity, precision tip, gel formula	1	~$5	Loctite Ultra Gel Control, 4 g
7	Inlay casting wax	Brown dental/jewelry sticky wax sticks, pliable at room temp	1 box	~$8	Brown Inlay Wax, 1 oz
8	BNC cables (male–male, 50 Ω)	1 m length, for Picoscope connection	2	~$10	Included with PicoScope kit; extras available at any electronics supplier
	Core materials (without scope)			**~38–53**	
	Core materials (with PicoScope)			**~230–245**	

Table D.2: Mitigation and Measurement Accessories

#	Component	Purpose (failure mode addressed)	Qty	Est. Cost	Source
9	Craft foam sheets (EVA, 6 mm thick)	Vibration-isolating cradle (FM 1: anchor loss)	1 pack	~$6	Any craft store or Amazon
10	Digital thermometer, 0.1 °C resolution	Thermal drift monitoring (FM 3)	1	~$12	Any kitchen/lab thermometer with 0.1 °C readout
11	Styrofoam cooler, small (~6-qt)	Thermal enclosure (FM 3: drift isolation)	1	~$5	Grocery or hardware store
12	Milligram precision scale (0.001 g)	Weighing wax perturbation masses	1	~$20	Amazon search: "milligram scale 0.001g"
13	Metric ruler, mm scale	Positioning masses along rod	1	~$3	Any school supply
14	Isopropyl alcohol, 91 %+	Cleaning glass rods before assembly	1 bottle	~$4	Any pharmacy
15	Safety glasses	Eye protection (glass rods can snap)	1 per person	~$3	Any hardware store
16	Masking tape	PZT alignment jig (FM 2: centering)	1 roll	~$4	Any hardware store
17	Fine-tip permanent marker	Marking positions on rod	1	~$2	Any office supply
18	Soft lint-free cloth	Handling and cleaning glass	several	~$2	Any store
19	Acetone (nail polish remover)	Emergency super-glue skin-bond release	1 small bottle	~$3	Any pharmacy
20	Plastic transfer pipettes (3 mL)	Placing water drops for Exp. 8 (rewritability)	1 pack	~$4	Amazon search: "plastic transfer pipettes"
21	Small bowl of water	CW excitation via wet finger (Exp. 7)	—	free	Tap water
	Accessories subtotal			**~$68**	
	Grand total (without scope)			**~106–121**	
	Grand total (with PicoScope)			**~298–313**	

> **Budget note.** Most schools already own an oscilloscope with FFT capability and a function generator—if so, skip item 5 and the core materials cost is just ~$38.For labs without a scope, we recommend the PicoScope 2204A($ 192), which provides both the waveform generator (transmit) and the digitizer (receive) in one USB device with free cross-platform software (PS7). Any oscilloscope with ≥200 kHz bandwidth and a separate function generator will also work. The 15 glass rods and 15 PZT discs provide enough spares for multiple student groups, breakage, and control experiments. One kit serves an entire class.

D.2 Safety Notes

⚠️ **Glass hazard.** Borosilicate rods can snap if bent or dropped. Always wear safety glasses when handling rods. Dispose of broken glass in a rigid sharps container, never a trash bag.

⚠️ **Cyanoacrylate (super glue).** Bonds skin instantly. Keep acetone (nail polish remover) on hand to dissolve accidental skin bonds. Work in a ventilated area. Supervise younger students closely during the gluing step.

⚠️ **Hearing.** The fundamental mode (17.7 kHz) is near the upper limit of human hearing. Some students may hear a faint whine during high-amplitude excitation. This is harmless at the drive levels used here (\<1 V), but if anyone reports discomfort, reduce the AWG amplitude to 0.1 V.

D.3 Experiment 1 — Building the Resonator

Objective: Assemble a functioning glass-rod acoustic resonator and verify electrical connectivity.

Time: 30 minutes active + 24 hours glue cure.

Materials: Items 1 or 2, 3 or 4, 6, 9, 16, 18 from the BOM.

Procedure:

1. **Clean the rod.** Wipe one glass rod with isopropyl alcohol and a soft cloth. Allow 2 minutes to dry completely. Finger oils dampen vibrations measurably.
2. **Build the foam cradle at the displacement nodes.** Cut two V-shaped notches in a strip of craft foam, spaced 75 mm apart, centered on the rod—this places each notch at $L/4$ and $3L/4$ from one end (37.5 mm and 112.5 mm for a 150 mm rod). These positions are the exact displacement nodes of the second longitudinal mode—the acoustic "stems" of the rod. The rod should rest horizontally in the notches with no hard contact. A wine glass rings because you hold it by the stem, a vibrational node where energy cannot escape; the same physics governs your rod mount (see Failure Mode 6 and Section 7).
3. **Center the PZT disc (critical).** Cut two small strips of masking tape (~12 mm each). Adhere them in a cross-hair pattern centered on the flat end face of the rod. The intersection marks the center of the 6 mm face—this is where the PZT must go. An off-center disc excites transverse and torsional modes that pollute the spectrum (Failure Mode 2).
4. **Apply glue sparingly.** Place one tiny drop of cyanoacrylate—smaller than a pinhead, less than 0.5 mm in diameter—at the center of the cross-hair. *Less is more:* excess glue adds mass and viscoelastic damping that destroys the quality factor (Failure Mode 1).
5. **Attach the PZT disc.** Press the flat face of the PZT disc onto the glued spot, centering it on the cross-hair. Hold firm, even pressure for 30 seconds. Gently peel away the tape cross-hair strips.
6. **Cure.** Set the assembly aside on the foam cradle for a full 24 hours. Cyanoacrylate reaches full bond strength overnight; rushing produces a weak, lossy joint.
7. **Connectivity check.** Connect the PZT leads to the PicoScope Channel A input via a BNC adapter or clip leads. Set the scope to AC coupling, 1 mV/div, 1 ms/div timebase. Gently tap the free end of the rod with a fingernail. You should see a decaying burst of oscillation on screen. If no signal appears: check wire connections, ensure the PZT is not cracked, and try a firmer tap.

> ⚙️ **Failure Mode 1 — PZT/Epoxy Boundary Damping (Low Q)**
>
> The glue joint and PZT disc add mass and internal friction at the rod end, potentially reducing Q from the intrinsic material value (~10,000) to below 100 if done carelessly.
>
> - Use the *absolute minimum* glue: one drop smaller than a pinhead.
> - Ensure the cured glue layer is thin (\<0.1 mm). A thick glue layer acts as a lossy viscoelastic coupling that absorbs acoustic energy each cycle.
> - Build a second "control" rod with *no* PZT attached—excite it by tap only and record its ring-down via a nearby microphone or laser vibrometer (if available). Comparing Q values between the two rods isolates the damping contribution of the PZT joint.

- If Q is unacceptably low (\<500 after Experiment 2), scrape off the PZT with a razor blade, clean the end face with alcohol, and re-glue with a smaller drop.

⚙️ **Failure Mode 6 — Support-Point Energy Drain (the Glass Harp Lesson)**

A wine glass rings brilliantly because you hold it by the stem—a point of zero vibrational displacement (a node) for the rim's flexural modes. Energy cannot leak through a point that isn't moving. Press your finger against the rim itself and the tone dies instantly: the support is now at an antinode, and your finger becomes an acoustic drain.

The same physics governs your rod mount. For a free-free rod vibrating in its n-th longitudinal mode, the displacement pattern is $u(x) = \cos(n\pi x/L)$. The displacement nodes—the "stems"—occur at:

Mode	Node positions
$n = 1$	$L/2$ (center)
$n = 2$	$L/4$ and $3L/4$
$n = 3$	$L/6$, $L/2$, and $5L/6$

The recommended cradle positions at $L/4$ and $3L/4$ (step 2) are the exact nodes of mode 2, and they produce only 50% of peak displacement for mode 1—a good all-round compromise for multi-mode measurements.

If your Q values are lower than expected despite a clean PZT bond:

- **Verify cradle positions.** Measure from one end: 37.5 mm and 112.5 mm for a 150 mm rod. Mark the positions with a fine-tip marker.
- **Try midpoint mounting.** For mode-1-only measurements, a single support at $L/2 = 75$ mm is the ideal node and will maximize Q for the fundamental. (This sacrifices mode 2, which has an antinode there.)
- **Upgrade to fishing line.** Loop a thin monofilament line around the rod at the node position and tension it between two fixed posts. This gives near-zero contact area and mimics the knife-edge mounts used in precision metrology. Students who have played a glass harp will immediately feel the analogy: the taut line is the stem.
- **Diagnostic test.** Slide the foam cradle to the rod's center ($L/2$) and measure Q for mode 1. Then slide it to $L/4$ and remeasure. If Q changes by more than 20%, your foam is too stiff or making hard contact—switch to softer foam, adjust the V-notch depth, or use fishing line.

D.4 Experiment 2 — Measuring the Quality Factor

Objective: Determine Q via two independent methods (ring-down and bandwidth) and confirm the resonator is material-loss-limited.

Time: 45 minutes.

Materials: Assembled resonator from Exp. 1, PicoScope 2204A, BNC cable, foam cradle.

Procedure:

1. **Set up.** Place the resonator on the foam cradle on a stable surface (no vibration from HVAC, foot traffic, etc.). Connect the PZT leads to both the PicoScope AWG output (waveform generator) and Channel A input. A BNC T-connector works, or simply share the two PZT leads between the AWG and scope clips.

2. **Impulse excitation.** In the PicoScope software (PS7, free download), configure the AWG to output a single 1-cycle burst at 17,700 Hz with 0.5 V amplitude. Set Channel A to trigger on the rising edge. Set the timebase to 50 ms/div so the display shows ~500 ms of data.
3. **Capture the ring-down.** Trigger the burst and record the decaying sinusoidal waveform. The oscillation frequency should be near 17.7 kHz; the amplitude should decay smoothly.
4. **Measure τ (time constant).** The envelope follows $A(t) = A_0\, e^{-t/\tau}$. Identify the initial peak amplitude A_0, then find the time t at which the envelope has dropped to $A_0/e \approx 0.368 \times A_0$. That time is τ. Alternatively, use PicoScope's cursor measurements to read the 1/e point directly.
5. **Calculate Q from ring-down:**

$$Q_{\text{ringdown}} = \pi\, f_1\, \tau$$

For example: $f_1 = 17{,}700$ Hz, $\tau = 180$ ms gives $Q = \pi \times 17{,}700 \times 0.180 = 10{,}006$.

1. **Cross-check via bandwidth method.** Switch PicoScope to spectrum (FFT) mode. Drive the rod with a slow linear frequency sweep (chirp) from 17,000 Hz to 18,500 Hz over 2 seconds at 0.2 V amplitude. In the FFT view, locate the resonance peak. Measure the full width at half-maximum power (the −3 dB bandwidth, $\Delta f_{3\text{dB}}$—the frequency span where the peak drops to 70.7% of its maximum amplitude, or −3 dB from the peak). Calculate:

$$Q_{\text{bandwidth}} = \frac{f_1}{\Delta f_{3\text{dB}}}$$

1. **Compare the two Q values.** They should agree within 10%. Record all results in Worksheet D.1.

> ⚙️ **Failure Mode 4 — Piezoelectric Self-Resonance Masking**
>
> A 10 mm PZT disc has its own radial resonance, typically at 200–300 kHz—well above the glass rod's fundamental (17.7 kHz). However, some disc types have a thickness resonance at lower frequencies that can overlap with glass-rod harmonics.
>
> To identify PZT artifacts:
>
> - *Before* gluing (during your next build), hold a bare PZT disc by its leads in free air and tap it gently. Record the FFT spectrum—these are PZT-only resonances.
> - After assembly, compare the full spectrum against the PZT-only spectrum. Any peak that appears in both is a PZT artifact, not a glass eigenmode.
> - True glass-rod longitudinal modes are evenly spaced at $\Delta f = v_{\text{bar}}/(2L) \approx 17{,}700$ Hz. PZT resonances do *not* follow this comb pattern. Any peak that falls off the comb by more than 1% is suspect.

Worksheet D.1 — Quality Factor Measurement

CWM Macro-Scale Experiment Guide · Coherent Wave Memory

Print one copy per rod. Record ring-down and bandwidth Q values for comparison.

Parameter	Rod 1	Rod 2	Rod 3
Date / Time	15 Mar 2026, 14:30		
Experimenter	A. Student		
Room temperature (°C)	22.3		
Rod length L (mm)	150.2		
Rod diameter d (mm)	6.01		
Glue amount (tiny / small / medium)	tiny		
Measured f_1 (Hz)	17,693		
Predicted $f_1 = 5{,}315/(2L)$ (Hz)	17,694		
Ring-down τ (ms)	168		
Q (ring-down) = $\pi f_1 \tau$	9,335		
−3 dB bandwidth Δf_{3dB} (Hz)	1.9		
Q (bandwidth) = $f_1/\Delta f_{3dB}$	9,312		
Two methods agree within 10%? (Y/N)	Y ✓		
Notes	Clean bond, thin glue		

Expected results. Q should fall in the range 1,000–10,000. Values near 10,000 indicate excellent construction (material-loss-limited). Values below 500 suggest excessive glue, poor PZT centering, or a hard contact point on the cradle—rebuild with corrections per the Failure Mode 1 mitigation above. A control rod with no PZT (tap-excited) should yield Q near 10,000, confirming the intrinsic glass value.

D.5 Experiment 3 — Mapping the Mode Spectrum

Objective: Identify the longitudinal eigenmode comb and distinguish real modes from transverse/torsional artifacts.

Time: 45 minutes.

Materials: Assembled resonator, PicoScope, BNC cable, foam cradle.

Procedure:

1. **Configure the chirp.** Set the PicoScope AWG to output a linear frequency sweep from 1 kHz to 200 kHz over 1 second at 0.5 V amplitude.
2. **Capture the response.** Record Channel A during the chirp with FFT mode enabled. Use a Hanning window and at least 16,384 FFT points for ~1 Hz frequency resolution.
3. **Identify the mode comb.** Longitudinal modes appear as evenly spaced peaks at:

Mode n	Predicted frequency
1	17,717 Hz
2	35,434 Hz
3	53,150 Hz
4	70,867 Hz
5	88,584 Hz
6	106,301 Hz
7	124,018 Hz
n	$n \times 5{,}315/(2L)$ Hz

1. **Mark confirmed modes.** On the FFT display (or in a screenshot), mark each peak that falls within ±0.5% of a predicted comb frequency. These are your confirmed longitudinal eigenmodes.
2. **Catalog spurious peaks.** Any peaks that do *not* fall on the comb are either: (a) transverse or torsional modes (Failure Mode 2), (b) PZT self-resonances (Failure Mode 4), or (c) electrical noise (harmonics of 60 Hz mains). Note their frequencies.
3. **Count confirmed modes** above the noise floor and record in Worksheet D.2.

> ⚙️ **Failure Mode 2 — Transverse/Torsional Spectrum Pollution**
>
> If the PZT disc is not centered on the rod axis, it applies an off-axis force that excites bending and torsional modes in addition to the desired longitudinal modes. These appear as extra peaks *between* the longitudinal comb lines.
>
> Mitigations:
>
> - **Verify centering.** If spurious peaks are within 10 dB of the longitudinal peaks, remove the PZT and re-glue with better centering (see Exp. 1, step 3).
> - **Software filter.** Export the FFT data (PicoScope can save .csv files) and discard all peaks more than ±500 Hz from the predicted comb frequencies.
> - **Rotation test.** Rotate the rod 90° on the foam cradle and re-measure. Longitudinal modes (which depend only on length) will not shift. Transverse modes (which depend on the rod's cross-sectional geometry) will shift because the cross-section is never perfectly circular. Any peak that moves under rotation is not longitudinal.

Worksheet D.2 — Mode Spectrum Map

CWM Macro-Scale Experiment Guide · Coherent Wave Memory

Print one copy per rod. Log predicted vs. measured frequencies for all confirmed longitudinal modes.

Mode n	Predicted f_n (Hz)	Measured f_n (Hz)	Δf = meas − pred (Hz)	Amplitude (dB)	Confirmed?
1	17,717	17,693	−24	−22	Y
2	35,434	35,388	−46	−28	Y
3	53,150	53,081	−69	−35	Y
4	70,867	70,775	−92	−43	Y
5	88,584	88,466	−118	−52	Y
6	106,301	106,160	−141	−63	Y
7	124,018				Y / N
Spurious 1	—	24,330	—	−58	type: transverse
Spurious 2	—		—		type: ___
Spurious 3	—		—		type: ___
Total confirmed longitudinal modes:		6			

Expected results. 5–10 clean longitudinal modes should be visible within the PicoScope's 10 MHz bandwidth. The first 2–3 modes will have the highest amplitude. Measured frequencies should match predictions within ±0.5%. If your rod is 200 mm instead of 150 mm, recalculate: $f_1 = 5{,}315/(2 \times 0.200) = 13{,}288$ Hz.

D.6 Experiment 4 — Thermal Stability Characterization

Objective: Quantify how room-temperature fluctuations shift mode frequencies, and establish a stable measurement protocol for perturbation experiments.

Time: 90 minutes (mostly waiting for equilibration).

Materials: Assembled resonator, PicoScope, digital thermometer, styrofoam cooler, foam cradle.

Background. Borosilicate glass has a thermal expansion coefficient of ~3.3×10^{-6} /°C and a temperature coefficient of elastic modulus of ~−100 ppm/°C. The net frequency sensitivity is approximately:

$$\frac{\Delta f}{f} \approx -50 \text{ ppm}/^\circ\text{C}$$

For $f_1 = 17{,}700$ Hz, this predicts $\Delta f \approx -0.9$ Hz per °C of temperature change.

Procedure:

1. **Open-air baseline.** Place the resonator on the foam cradle on a lab bench. Place the thermometer probe within 2 cm of the rod. Every 5 minutes for 30 minutes, record f_1 (using the bandwidth method from Exp. 2—a quick FFT snapshot) and the temperature. Do not touch the setup or breathe on it.

2. **Thermal perturbation.** Breathe warm air directly onto the rod for 10 seconds, or place a cup of warm water (~50 °C) 20 cm away. Resume recording f_1 and temperature every 2 minutes until the frequency returns to within 0.5 Hz of its initial value. Note the recovery time.
3. **Insulated baseline.** Place the entire resonator assembly (foam cradle, rod, PZT, and thermometer probe) inside the styrofoam cooler. Thread the BNC cable and thermometer wire through a small notch cut in the cooler lid. Close the lid. Repeat the 30-minute baseline: record f_1 and temperature every 5 minutes.
4. **Calculate the temperature coefficient.** Plot f_1 versus temperature for both the open-air and insulated datasets. The slope $\Delta f/\Delta T$ is your measured temperature coefficient. Compare to the predicted -0.9 Hz$/$°C.
5. **Record results** in Worksheet D.3.

> ⚙️ **Failure Mode 3 — Thermal Drift Swamping Perturbation Encoding**
>
> A 1 °C temperature change shifts modes by ~0.9 Hz—comparable to a small wax perturbation. If thermal drift is not controlled, perturbation signals drown in noise.
>
> Mitigations:
>
> - **Always use the styrofoam enclosure** for Experiments 5 and 6. This alone reduces drift by 10–50×.
> - **Equilibrate 30 minutes** before starting perturbation experiments.
> - **Use differential measurement.** Thermal drift shifts *all* modes by the same fractional amount (-50 ppm$/$°C). A localized mass perturbation shifts each mode by a *different* amount (proportional to $\sin^2(n\pi x/L)$). Track the *pattern* of relative shifts, not absolute frequencies. This is your strongest discriminant.
> - **Bracket measurements.** Take a "before" spectrum immediately before applying wax, and an "after" spectrum within 60 seconds. Over such short intervals, thermal drift is negligible.
> - **Record temperature at every measurement point** so post-hoc correction is possible.

Worksheet D.3 — Thermal Stability Log

CWM Macro-Scale Experiment Guide · Coherent Wave Memory

Print one copy per thermal stability run. Track frequency drift vs. temperature over time.

Time (min)	Temp (°C)	f_1 (Hz)	Δf from t = 0 (Hz)	Environment
0	22.3	17,693.2	0.0	Insulated
5	22.3	17,693.1	−0.1	Insulated
10	22.4	17,693.0	−0.2	Insulated
15	22.4	17,692.9	−0.3	Insulated
20	22.4	17,693.0	−0.2	Insulated
25	22.4	17,693.0	−0.2	Insulated
30	22.4	17,693.0	−0.2	Insulated
(thermal perturbation applied)	hand near enclosure 30 s			
32	22.8	17,692.6	−0.6	Open
34	23.1	17,692.3	−0.9	Open
36	23.0	17,692.4	−0.8	Open
38	22.8	17,692.6	−0.6	Open
40	22.6	17,692.8	−0.4	Open
Measured $\Delta f/\Delta T$:		−0.85 Hz/°C		
Predicted (−0.9 Hz/°C):		agrees within 6%		
Recovery time after perturbation:		~8 min		

Expected results. The open-air drift rate will depend on your HVAC system—typically 0.5–5 Hz over 30 minutes. Inside the styrofoam enclosure, drift should be \<0.5 Hz over 30 minutes, which is small enough to resolve wax perturbation shifts of 1–10 Hz.

D.7 Experiment 5 — Perturbation Encoding

Objective: Write data to the rod by applying wax masses and verify that measured frequency shifts match Rayleigh perturbation theory.

Time: 60 minutes.

Materials: Assembled resonator (inside styrofoam enclosure), PicoScope, wax, milligram scale, ruler, fine-tip marker.

Background. The Rayleigh perturbation formula predicts that a small mass δm placed at position x along a rod of total mass M and length L shifts the n-th mode frequency by:

$$\frac{\Delta f_n}{f_n} = -\frac{\delta m}{2M}\sin^2\left(\frac{n\pi x}{L}\right)$$

The key insight is that *different modes shift by different amounts* depending on where the mass sits relative to each mode's antinodes. This position-dependent pattern of shifts is the "spectral fingerprint"—the basis of data encoding.

Procedure:

1. **Prepare wax pellets.** Pinch off small pieces of inlay wax and roll them into balls approximately 1 mm in diameter. Weigh each on the milligram scale and record the mass (target: 0.05–0.5 mg). Prepare 3–5 pellets of varied sizes.
2. **Mark positions.** Using the ruler and fine-tip marker, mark reference positions on the rod at 25%, 33%, 50%, 67%, and 75% of the rod length from the PZT end (i.e., at 37.5, 50.0, 75.0, 100.0, and 112.5 mm for a 150 mm rod).
3. **Record the unperturbed spectrum.** Drive the rod with a chirp and capture the FFT. Record the frequencies of modes 1–5 in Worksheet D.4 under "Before."
4. **Apply one wax pellet.** Press it onto the rod at a marked position x. The wax should adhere by its own stickiness —no glue needed. Record the pellet mass and position.
5. **Record the perturbed spectrum.** Immediately re-drive and capture the FFT. Record mode frequencies under "After." Calculate the shift Δf_n for each mode.
6. **Compare to theory.** Calculate the predicted $\Delta f_n / f_n$ for each mode using the Rayleigh formula above. Weigh the rod itself to get M (typically ~6 g for a 150 mm × 6 mm borosilicate rod, density 2,230 $\mathrm{kg/m^3}$).
7. **Repeat with different positions.** Move the wax to a different marked position and re-measure. The fingerprint should change: modes whose antinodes are near the new wax position shift most; modes with nodes near the wax barely shift.
8. **Verify reversibility.** Remove the wax with a fingernail and re-measure the spectrum. Frequencies should return to the unperturbed values within the thermal drift tolerance established in Experiment 4.

> ⚙️ **Failure Mode 5 — Non-Linear Acoustic Coupling**
>
> At high drive amplitudes, acoustic energy leaks between modes via non-linear elastic coupling, smearing the spectral fingerprint and reducing discrimination.
>
> Mitigations:
>
> - **Keep drive amplitude ≤ 0.5 Vpp.** This keeps the rod well within the linear regime.
> - **Verify linearity.** Measure mode amplitudes at 0.25 V, 0.50 V, and 1.00 V. Plot amplitude vs. drive voltage. The relationship should be linear (doubling voltage doubles response amplitude). If doubling drive voltage increases any mode's amplitude by more than 2.2× (exceeding the linear 2.0× by more than 10%), you have entered the non-linear regime—reduce the drive.
> - **Listen for audible ringing.** If you can clearly hear the rod vibrating (a high-pitched whine), the amplitude is too high. Reduce AWG output until the sound is barely perceptible or inaudible.

Worksheet D.4 — Perturbation Encoding Data

CWM Macro-Scale Experiment Guide · Coherent Wave Memory

Print one copy per perturbation trial. Record before/after mode frequencies and compare to Rayleigh predictions.

	Trial info
Trial #	1
Wax pellet mass δm (mg)	0.8
Position x (mm from PZT end)	75.0
Position x/L (fraction)	0.500
Rod mass M (g)	9.52
Rod length L (mm)	150.2
Temperature (°C)	22.4

Mode *n*	f_n before (Hz)	f_n after (Hz)	Δf_n measured (Hz)	$\Delta f_n/f_n$ meas. (ppm)	$\Delta f_n/f_n$ predicted (ppm)	Error (%)
1	17,693.2	17,692.5	−0.7	−40	−42	5
2	35,388.1	35,388.0	−0.1	−3	0	–
3	53,081.0	53,078.8	−2.2	−41	−42	2
4	70,775.3	70,775.2	−0.1	−1	0	–
5	88,466.1	88,462.4	−3.7	−42	−42	1

After removing wax:

Mode *n*	f_n recovered (Hz)	Δf from original (Hz)	Recovered within ±0.5 Hz?
1	17,693.1	−0.1	Y
2	35,388.0	−0.1	Y
3	53,080.9	−0.1	Y
4	70,775.2	−0.1	Y
5	88,466.0	−0.1	Y

Expected results. Mode shifts should match Rayleigh predictions within 5%. The mode whose antinode is nearest the wax position will shift most; modes with nodes near the wax will barely shift. Frequencies should recover fully after wax removal. If the match is poor (>10% error), verify that the wax mass is small compared to the rod mass ($\delta m/M \ll 1$) and that the temperature has not drifted during the measurement.

D.8 Experiment 6 — Associative Recall Demonstration

Objective: Show that the rod physically distinguishes between matching and non-matching query patterns—the basis of CWM's O(1) nearest-neighbor search.

Time: 60 minutes.

Materials: Assembled resonator, PicoScope, wax, ruler, foam cradle, styrofoam enclosure.

Procedure:

1. **Create Pattern A.** Apply two wax pellets at the quarter-points of the rod: $x = L/4 = 37.5$ mm and $x = 3L/4 = 112.5$ mm. Record the resulting spectral fingerprint—the frequencies of modes 1–5 in their shifted positions.
2. **Build the query signal.** In the PicoScope AWG, create a multi-tone waveform composed of the five shifted frequencies from Pattern A's fingerprint. Set each tone to equal amplitude (0.1 V per tone). This is "Query A."
3. **Measure the matched response.** Drive the rod with Query A. The rod contains Pattern A, so the query *matches* the stored pattern. Record the peak response amplitude in dB from the FFT.
4. **Create Pattern B.** Remove Pattern A's wax. Apply two pellets at the third-points: $x = L/3 = 50$ mm and $x = 2L/3 = 100$ mm. Record Pattern B's fingerprint.
5. **Drive with Query A (mismatched).** Replay Query A—the waveform built from Pattern A's frequencies. The rod now contains Pattern B, so the query does *not* match. Record the peak response amplitude. It should be significantly lower than in step 3.
6. **Drive with Query B (matched).** Build a new multi-tone waveform from Pattern B's frequencies. Drive the rod with this Query B. The rod now matches, so the response should be strong. Record the amplitude.
7. **Calculate the discrimination margin.** The difference in dB between the matched response and the best non-matched response is the discrimination margin. A margin ≥15 dB means the correct match produces ≥30× more power than the closest incorrect match.
8. **Record results** in Worksheet D.5.

Worksheet D.5 — Associative Recall Discrimination

CWM Macro-Scale Experiment Guide · Coherent Wave Memory

Print one copy per discrimination test. Log response amplitudes for matched vs. unmatched query patterns.

Pattern stored on rod	Query driven	Peak response (dB)	Match?
Pattern A	Query A (matching)	−22	✓
Pattern B	Query A (non-matching)	−44	✗
Pattern B	Query B (matching)	−23	✓

Metric	Value
Discrimination margin (matched − non-matched):	21 dB
Power ratio (10^(margin/10)):	126×
Sufficient for reliable detection? (≥15 dB)	Y ✓

Expected results. The discrimination margin should be 15–25 dB, meaning the matching pattern produces 30–300× more acoustic power than a non-matching pattern. If the margin is below 10 dB, check that (a) the two wax patterns are placed at genuinely different positions, (b) the drive amplitude is in the linear regime (≤0.5 V per tone), and (c) the query frequencies accurately match the stored pattern's measured frequencies.

D.9 Experiment 7 — CW Precision Readout (Bowing vs. Ringing)

Objective: Demonstrate that continuous-wave (CW) excitation with lock-in detection yields higher SNR than impulse readout—and experience the difference physically by "bowing" the rod with a wet finger, just as a glass harmonica is played.

Time: 60 minutes.

Materials: Assembled resonator, PicoScope, BNC cable, foam cradle, styrofoam enclosure, small bowl of water.

Background. Experiments 1–6 use impulse readout: strike the rod, listen to it ring down, measure the FFT. This is fast (one ringdown time $\tau \approx 180$ ms) but limited in SNR—the measurement window is fixed at τ, and all the noise within the bandwidth $\sim 1/\tau$ contributes. An alternative is CW readout: drive the rod continuously at a single mode frequency and measure the steady-state response. A lock-in detection technique—multiplying the response by the drive signal and averaging—rejects all noise outside a narrow bandwidth $\sim 1/(2T_{\text{int}})$. The SNR gain over impulse is:

$$\text{Gain (dB)} = 10 \log_{10}\left(\frac{T_{\text{int}}}{\tau}\right)$$

At $T_{\text{int}} = 1$ s: +7.5 dB. At $T_{\text{int}} = 10$ s: +17.5 dB. At $T_{\text{int}} = 60$ s: +25.2 dB (see Figure 15). This is the same physics that makes a glass harmonica so expressive: the performer's sustained finger contact provides continuous energy input, allowing the bowl to reach full amplitude and sustain it indefinitely—something a single tap cannot do.

Part A: Electronic CW Readout

1. **Set the CW drive.** Configure the PicoScope AWG to output a *continuous* sine wave at your rod's measured f_1 from Experiment 2. Set amplitude to 0.2 V. Let it run—do not stop it.

2. **Wait for ring-up.** The rod's amplitude builds exponentially with the same time constant τ as the ring-down. After $3\tau \approx 540$ ms, the rod has reached 95% of its steady-state amplitude. Wait at least 1 second.
3. **Capture a 1-second record.** In PicoScope, set the timebase to capture exactly 1 second of Channel A data. Save the raw waveform (.csv export).
4. **Software lock-in detection.** In a spreadsheet or Python script, perform the lock-in calculation:
5. Generate a reference signal: $\text{ref}(t) = \sin(2\pi f_1 t)$.
6. Multiply the captured signal by the reference: $\text{product}(t) = \text{signal}(t) \times \text{ref}(t)$.
7. Average the product over the 1-second window. This is the lock-in output—proportional to the mode amplitude, with all out-of-band noise rejected.
8. Record the lock-in amplitude as $A_{\text{CW,1s}}$.
9. **Repeat with 10-second capture.** Extend the AWG run and capture 10 seconds. Repeat the lock-in calculation. Record $A_{\text{CW,10s}}$.
10. **Compare to impulse.** From Experiment 2, you have the impulse ring-down peak amplitude A_{impulse}. Calculate the SNR improvement:

$$\text{Gain (dB)} = 20\log_{10}\left(\frac{A_{\text{CW}}}{A_{\text{impulse}}}\right)$$

Expected: ~7.5 dB for 1-second CW, ~17.5 dB for 10-second CW.

Part B: Bowing the Rod (the Glass Harmonica Experiment)

This part requires no electronics—just your hands and ears (plus the PZT to record what happens).

1. **Wet your finger.** Dip one finger in the bowl of water. You want a thin, even film of moisture—not dripping wet.
2. **Bow the rod.** With the PicoScope recording on Channel A (long timebase, ~5 seconds), run your wet finger firmly and steadily along the length of the rod, maintaining even pressure. The finger creates stick-slip friction that excites the rod's longitudinal modes, exactly as a wet finger on a wineglass rim excites its radial modes. You should hear a faint singing tone and see a sustained oscillation on the PicoScope screen.
3. **Compare the waveform.** The bowed response should show sustained, roughly constant-amplitude oscillation for as long as your finger maintains contact—in contrast to the exponentially decaying impulse ring-down from Experiment 2. *This is the glass harmonica in action.*
4. **Measure bowed frequency.** Switch to FFT mode while bowing. The dominant peak should fall at or near f_1. The excitation is broadband (stick-slip friction contains many frequencies), so you may also see modes 2 and 3 responding.
5. **Record results** in Worksheet D.6.

Worksheet D.6 — CW Readout Comparison

CWM Macro-Scale Experiment Guide · Coherent Wave Memory

Print one copy per CW session. Compare impulse ring-down vs. continuous-wave lock-in readout.

Measurement	Value
Impulse ring-down peak amplitude A_{impulse} (mV)	12.3
CW lock-in amplitude, 1 s ($A_{\mathrm{CW,1s}}$) (mV)	28.4
CW lock-in amplitude, 10 s ($A_{\mathrm{CW,10s}}$) (mV)	91.2
Gain (1 s): 20·log$_{10}$($A_{\mathrm{CW,1s}}$ / A_{impulse})	7.3 dB
Gain (10 s): 20·log$_{10}$($A_{\mathrm{CW,10s}}$ / A_{impulse})	17.4 dB
Expected gain (1 s): 7.5 dB	Y ✓
Expected gain (10 s): 17.5 dB	Y ✓
Wet-finger bowing: sustained oscillation observed?	Y ✓
Bowed frequency (Hz)	17,694
Bowed duration (s)	3.2
Notes on bowing technique	Slow steady stroke, 2nd attempt best

Expected results. The electronic CW gains should match predictions within ±3 dB (measurement noise, PZT coupling variability, and frequency drift all widen the tolerance). The wet-finger bowing should produce sustained oscillation at f_1 lasting as long as the finger maintains contact—typically 2–5 seconds per stroke. Students who have played a glass harp will find this immediately intuitive.

> **Why this matters.** Experiments 1–6 demonstrate CWM as a *rung bell*—impulse excitation, finite-duration readout. This experiment demonstrates CWM as a *bowed string*—continuous excitation, arbitrarily long integration, progressively higher precision. The two-phase readout architecture proposed in §2.3 combines both: ring the bell (fast, broadband) to find the answer, then bow the string (slow, precise) to read it exactly.

D.10 Experiment 8 — Rewritable Encoding with Water Drops

Objective: Demonstrate rewritable spectral encoding using water drops as removable mass perturbations—the transition from glass harmonica (fixed tuning) to Franklin's armonica (reconfigurable).

Time: 45 minutes.

Materials: Assembled resonator, PicoScope, plastic transfer pipettes, water, foam cradle, styrofoam enclosure, ruler, fine-tip marker, digital thermometer.

Background. In a glass harmonica, each bowl's pitch is set by its geometry—grind it to a specific diameter and thickness, and the frequency is fixed forever. But adding water to a bowl lowers its pitch by increasing the effective vibrating mass. The performer cannot change the glass, but *can* change the water level. This is precisely the Rayleigh perturbation mechanism: water at position x shifts mode n by an amount proportional to $\sin^2(n\pi x/L)$. Unlike wax

(Experiment 5), water evaporates—making the perturbation *rewritable*. This experiment demonstrates the harmonica-to-armonica transition: write a pattern with water, read the shifted spectrum, let it evaporate (erase), write a different pattern, and confirm the spectrum changes.

Procedure:

1. **Prepare.** Place the resonator horizontally on the foam cradle inside the styrofoam enclosure. Equilibrate for 15 minutes with the thermometer monitoring temperature. Ensure the rod surface is clean and dry.
2. **Record the unperturbed spectrum (Pattern 0).** Chirp the rod and record the FFT. Log frequencies of modes 1–5.
3. **Write Pattern 1.** Using a transfer pipette, place one small water drop (~2 mm diameter, ~4 μL) at the rod midpoint ($x = L/2 = 75$ mm). The water should sit as a bead on the glass surface. Record the time.
4. **Read Pattern 1.** Immediately chirp the rod and record the FFT. Log the shifted mode frequencies. Note: mode 1 (which has an antinode at $L/2$) should shift the most. Mode 2 (which has a *node* at $L/2$) should shift the least—confirming position-dependent encoding.
5. **Erase Pattern 1.** Gently blot the water drop with a lint-free cloth, or simply wait 3–5 minutes for it to evaporate. Re-measure the spectrum and confirm frequencies return to within ±0.5 Hz of Pattern 0.
6. **Write Pattern 2.** Place a water drop at $x = L/4 = 37.5$ mm. Immediately read the spectrum. The pattern of mode shifts should be *different* from Pattern 1—mode 2 (antinode at $L/4$) now shifts strongly, while mode 1 shifts less than before.
7. **Erase Pattern 2.** Remove the water and confirm spectral recovery.
8. **Write Pattern 3 (two drops).** Place drops at both $L/4$ and $3L/4$. The resulting shift pattern is distinct from both Pattern 1 and Pattern 2.
9. **Record results** in Worksheet D.7.

Worksheet D.7 — Rewritable Water-Drop Encoding

CWM Macro-Scale Experiment Guide · Coherent Wave Memory

Print one copy per rewritability session. Track spectra across write / erase / rewrite cycles.

Mode n	f (Pattern 0)	f (Pattern 1: $L/2$)	Δf_1 (Hz)	f (Pattern 2: $L/4$)	Δf_2 (Hz)	f (Pattern 3: $L/4 + 3L/4$)	Δf_3 (Hz)
1	17,693.2	17,689.5	−3.7	17,691.3	−1.9	17,689.5	−3.7
2	35,388.1	35,388.0	−0.1	35,380.7	−7.4	35,373.2	−14.9
3	53,081.0	53,069.9	−11.1	53,075.4	−5.6	53,069.9	−11.1
4	70,775.3	70,775.2	−0.1	70,775.2	−0.1	70,775.2	−0.1
5	88,466.1	88,447.5	−18.6	88,456.8	−9.3	88,447.5	−18.6

Verification	Result
Pattern 0 recovered after erasing Pattern 1? (±0.5 Hz)	Y ✓
Pattern 0 recovered after erasing Pattern 2? (±0.5 Hz)	Y ✓
Pattern 1 and Pattern 2 are spectrally distinguishable?	Y ✓
Pattern 3 is distinct from both Pattern 1 and Pattern 2?	Y ✓
Total distinct patterns written and read in this experiment	3

Expected results. Water drops are lighter than wax pellets, so frequency shifts will be smaller—typically 0.2–2 Hz per mode, depending on drop volume. The thermal enclosure is essential here to keep drift below the perturbation signal. The key observation is that the three patterns produce three *different* spectral fingerprints, and that erasing (removing water) fully recovers the baseline—demonstrating the write/read/erase cycle. If shifts are too small to resolve, use larger drops or a more sensitive frequency measurement (CW lock-in from Experiment 7 provides higher precision).

> **The Franklin insight.** This experiment turns the glass rod into a primitive armonica. The glass itself never changes—only the water does. You have just demonstrated the separation principle described in §12.5: the rod is optimized for resonance quality (Q), and the "write medium" (water) is a separate, removable layer. Franklin would recognize this immediately: same glass, different tuning, reconfigurable.

D.11 — Consolidated Experiment Log

CWM Macro-Scale Experiment Guide · Coherent Wave Memory

Photocopy this page for each student group or session. Attach completed Worksheets D.1–D.7.

Field	Entry
Experimenter name(s)	
Date	
School / Institution	
Rod serial # (label each rod)	
Rod length L (mm)	
Rod diameter d (mm)	
Rod mass M (g)	
PZT disc serial #	
PicoScope model & serial	
Room temperature at start (°C)	
Relative humidity (%)	
Foam cradle material	
Thermal enclosure used? (Y/N)	
Experiments completed (circle)	1 2 3 4 5 6 7 8
Best Q measured	
Number of confirmed longitudinal modes	
Best discrimination margin (dB)	
CW lock-in gain at 10 s (dB)	
Wet-finger bowing successful? (Y/N)	
Water-drop patterns written & erased	
Anomalies or unexpected observations	

D.12 Troubleshooting Guide

Symptom	Likely cause	Fix
No signal when tapping rod	Broken PZT lead or bad BNC connection	Inspect and re-solder PZT leads; verify BNC cable continuity with a multimeter
Q \< 100	Excessive glue (thick viscoelastic layer)	Remove PZT, clean end face, re-glue with a smaller drop; cure 24 hours
Q \< 500	PZT too large, or rod touching a hard surface	Ensure foam cradle has soft V-notches with no metal or plastic contact; try 5 mm PZT disc
Many peaks between expected modes	Off-center PZT exciting transverse modes	Remove PZT, re-glue centered using tape cross-hair; or filter FFT to comb frequencies only
f_1 much higher or lower than 17,700 Hz	Rod length ≠ 150 mm	Recalculate: $f_1 = 5{,}315/(2L)$. A 200 mm rod gives f_1 = 13,288 Hz; a 125 mm rod gives f_1 = 21,260 Hz
Mode frequencies drifting over minutes	Temperature fluctuation	Use styrofoam enclosure; equilibrate 30 min; record temperature at each data point
Wax won't stick to rod	Surface too smooth, or wax too cold	Warm wax between fingers for 30 s; lightly scuff rod surface with 400-grit sandpaper at the attachment point
FFT shows 60/120/180 Hz peaks	Electrical mains noise pickup	Use shorter BNC cables; move away from power strips and monitors; ensure PicoScope is USB-powered (not near wall adapter)
Response identical for all query patterns	Drive amplitude in non-linear regime	Reduce AWG amplitude to ≤ 0.5 Vpp; verify linearity per Exp. 5 procedure
A strong peak appears that is not on the mode comb	PZT self-resonance	Compare against bare-PZT spectrum (see Exp. 2 FM 4 mitigation); flag and exclude from analysis
Rayleigh prediction error > 10%	Wax mass too large relative to rod, or position measurement inaccurate	Use smaller wax (\<0.1 mg); measure position to ±0.5 mm; weigh rod precisely
CW lock-in gain much less than expected	AWG frequency not precisely at mode peak; PZT coupling asymmetric	Fine-tune AWG frequency in 0.1 Hz steps while watching lock-in amplitude; ensure firm PZT bond
No sustained tone when bowing with wet finger	Finger too dry, too wet, or pressure too light	Re-wet finger to a thin film; apply firm, steady pressure; move slowly (1–2 cm/s); try rosined cloth as alternative
Water drop slides off rod immediately	Rod surface too smooth or tilted	Ensure rod is perfectly horizontal; use smaller drops (~2 mm); lightly breathe on the surface to create a condensation film
Water-drop frequency shifts too small to resolve	Drop too small relative to rod mass	Use larger drops (up to ~5 mm diameter); or use CW lock-in (Exp. 7) for higher frequency resolution

D.13 Contributing Your Data

We invite all experimenters—students, teachers, hobbyists, and researchers—to submit completed worksheets and raw PicoScope data files to the project repository. Community data from diverse rod lengths, diameters, glass types, and environments will strengthen the empirical foundation of CWM and accelerate the transition from macro prototype to MEMS fabrication.

To contribute:

1. Photograph or scan your completed Worksheets D.1–D.7 and the Experiment Log (D.11).
2. Export raw PicoScope waveform files (.psdata or .csv) for each experiment.

3. Submit via pull request to the project repository at github.com/miketierce/wcfoma in the `data/community/` directory. Include your Experiment Log as the commit message or PR description.
4. Alternatively, email data files and scanned worksheets to the corresponding author.

Every data point matters. A middle school classroom measuring Q = 3,000 on a 200 mm rod constrains the same physics as a university lab measuring Q = 9,000 on a 150 mm rod—and the diversity of rod geometries and construction techniques is itself scientifically valuable. Unexpected results (low Q, spurious modes, anomalous thermal coefficients) are *especially* welcome: they reveal failure modes that improve the guide for the next experimenter.

All quantitative claims computed from first-principles simulation code (34 modules, 1,036 automated tests, all passing) and independently validated by finite element analysis. No curve fitting, no adjusted parameters, no post-hoc corrections. Repository: github.com/miketierce/wcfoma.

Illustration Plates

The following pages present each figure at full landscape scale for detailed examination. Pages are arranged for double-sided printing: each illustration appears on the front of its own sheet.

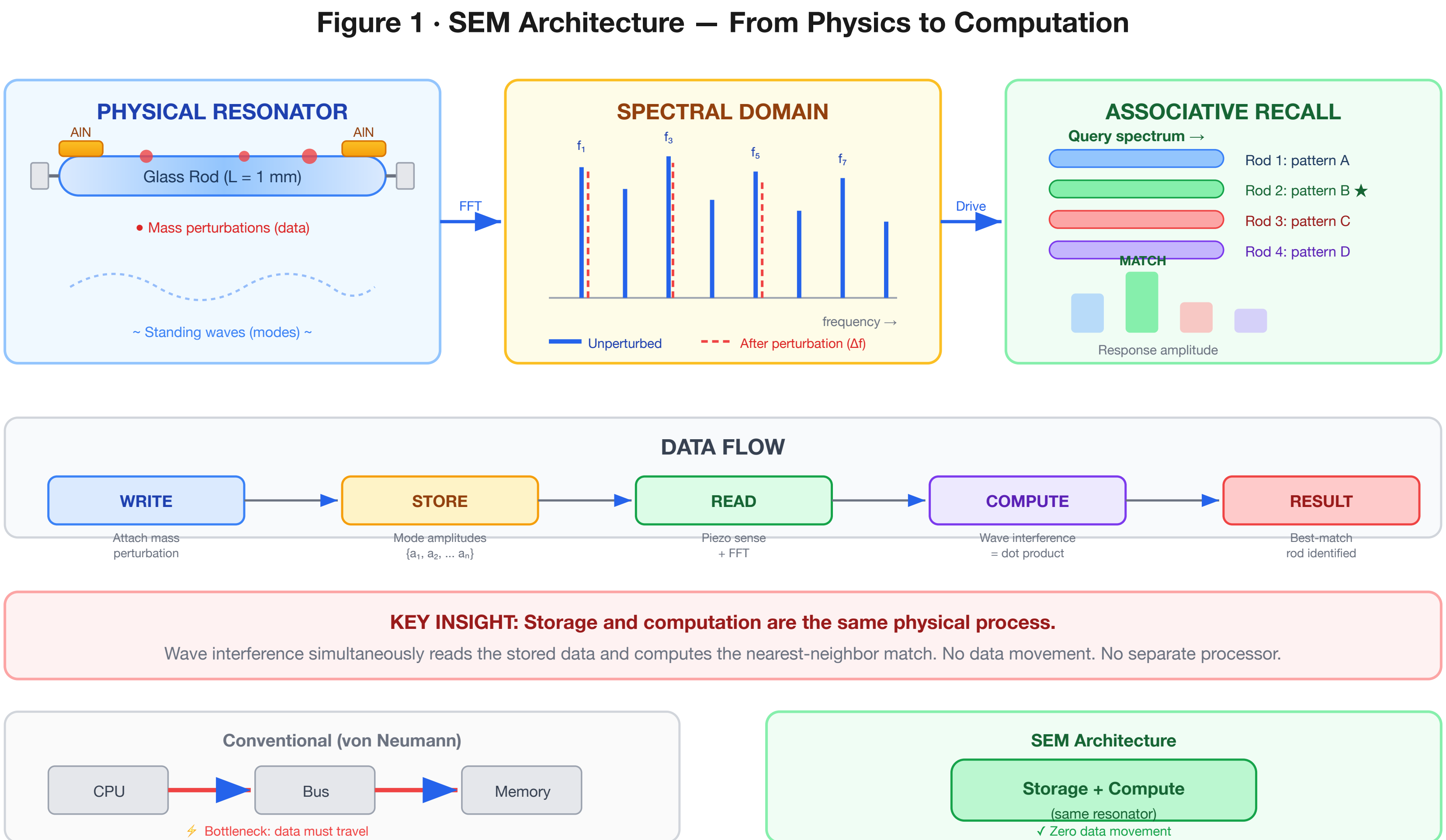


Figure 1. CWM architecture overview. *Left:* A glass resonator stores data as mass perturbations that shift eigenmode frequencies. *Center:* The spectral domain shows mode peaks before (solid) and after (dashed) perturbation — each pattern is a unique spectral fingerprint. *Right:* Driving an array with a query spectrum, the best-matching rod produces the largest response — a physical O(1) nearest-neighbor search. *Bottom:* Unlike von Neumann architectures, CWM unifies storage and computation in the same physical process.

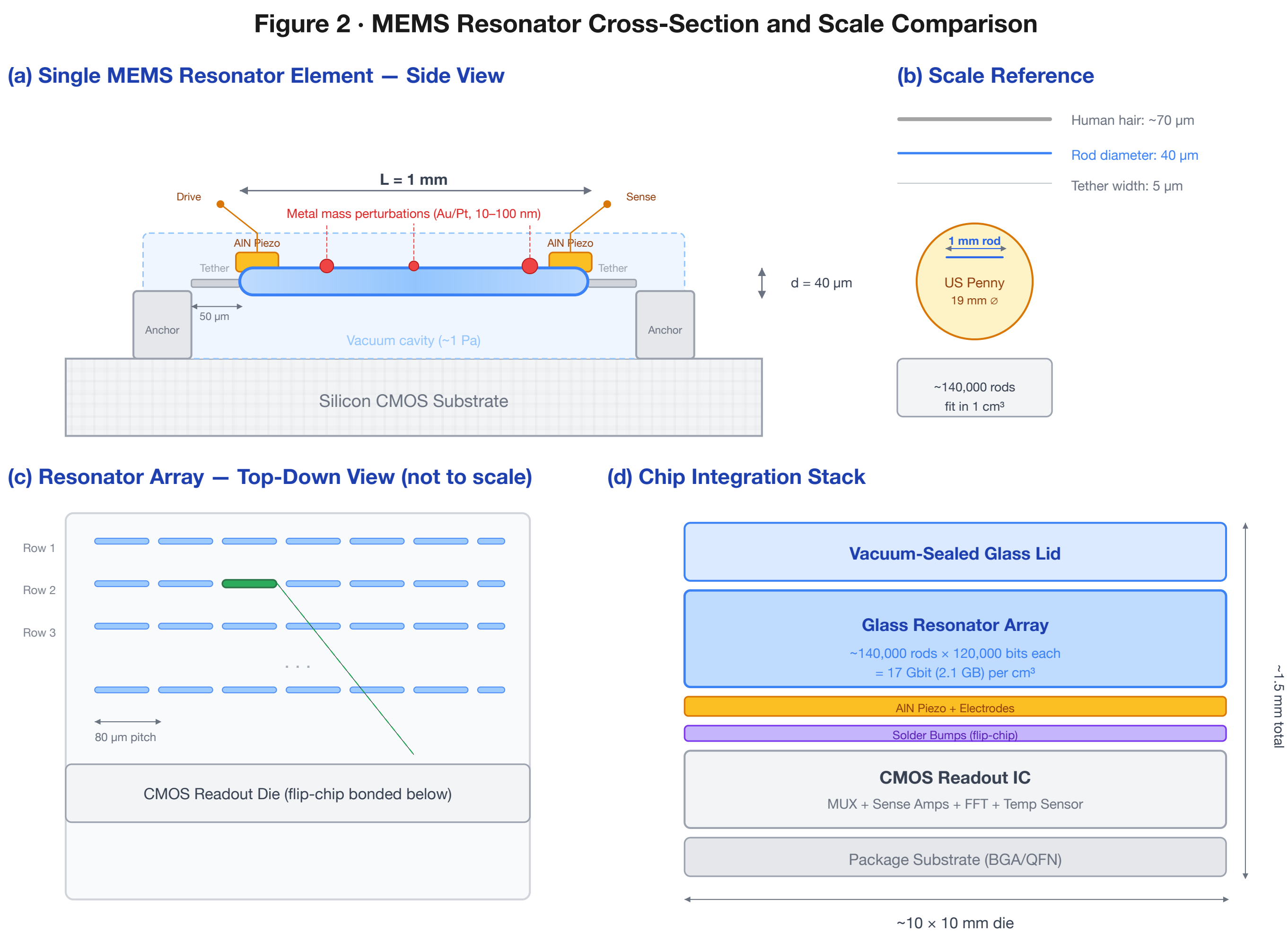


Figure 2. MEMS resonator at chip scale. *(a)* Cross-section of a single 1 mm × 40 μm glass resonator element showing AlN piezo transducers, anchor tethers, vacuum cavity, and lithographic mass perturbations. *(b)* Scale comparison: the rod diameter (40 μm) is thinner than a human hair (70 μm). *(c)* Top-down view of a resonator array at 80 μm pitch on a CMOS readout die. *(d)* Chip integration stack: vacuum-sealed glass array flip-chip bonded to CMOS — identical to existing MEMS accelerometer packaging.

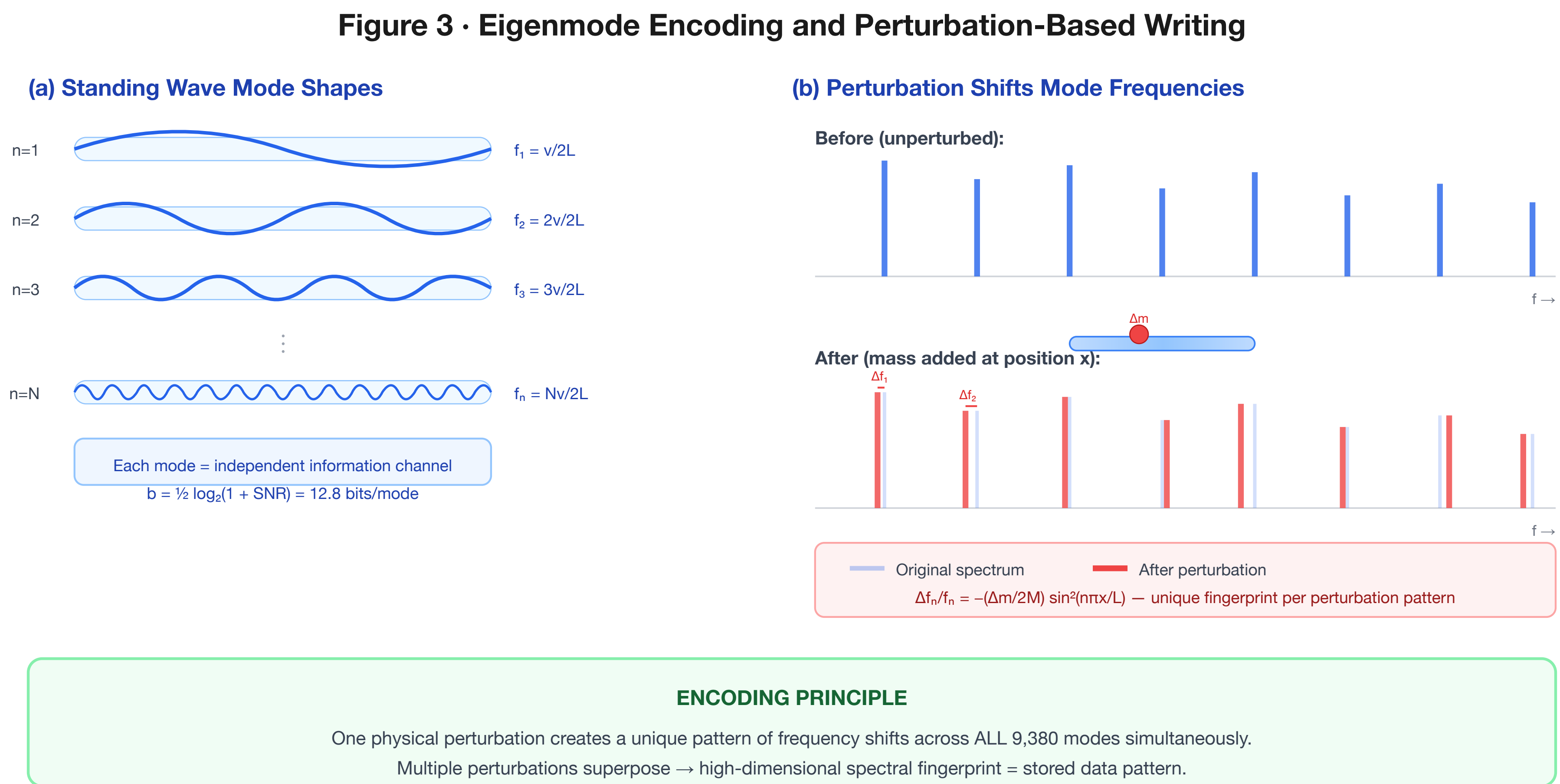


Figure 3. Eigenmode encoding. *(a)* Standing wave mode shapes for modes n = 1, 2, 3, … N within a glass rod — each mode is an independent information channel at a distinct frequency. *(b)* Perturbation effect: adding mass at a specific position shifts each mode frequency by a different amount (Rayleigh perturbation formula), creating a unique spectral fingerprint that encodes the perturbation's position and mass.

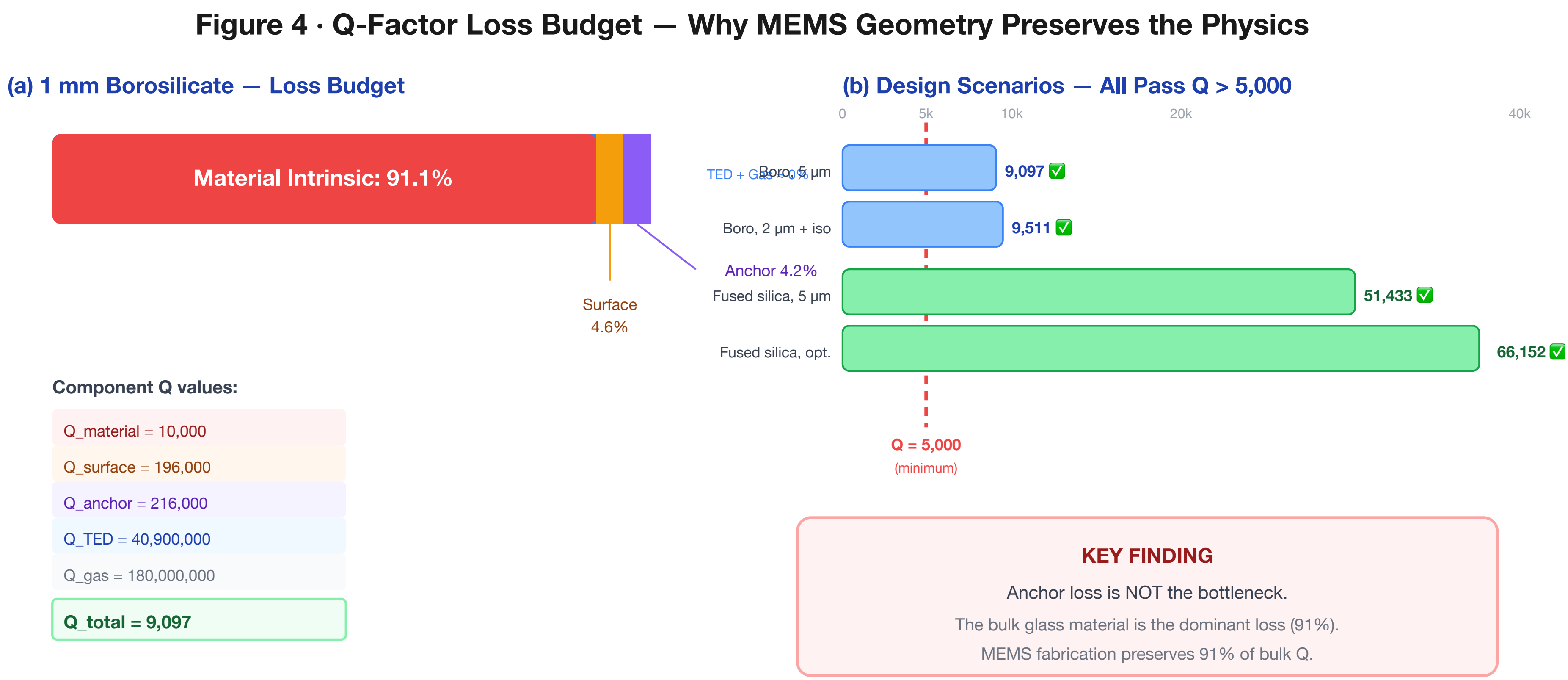


Figure 4. MEMS Q-factor loss budget. *(a)* For the reference 1 mm borosilicate design, material intrinsic loss dominates at 91.0% — anchor loss is only 4.4% of the total budget. $Q_{total} = 9{,}097$. *(b)* All four design scenarios (varying glass type, tether geometry, and vacuum level) clear the $Q > 5{,}000$ threshold for competitive operation, with fused silica designs exceeding $Q = 50{,}000$.

Figure 5 · Scaling: Same Mode Count, Exploding Density

Figure 5. Scaling from macro to MEMS. *(a)* Size comparison of the 150 mm prototype rod (0.04 Mbit/cm³), the 1 mm borosilicate MEMS target (95.1 Gbit/cm³ active, 9.5× DRAM), and a 0.5 mm fused silica array design (1.4 Tbit/cm³ packed-array, 1.4× NAND Flash). All designs share the same thermally stable mode physics; density explodes because volume shrinks as L^3 while capacity falls only as log L. *(b)* Log–log density plot showing CWM crossing DRAM at 2.1 mm, PCM at 1.15 mm, and NAND Flash at 0.45 mm — all within standard MEMS fabrication range.

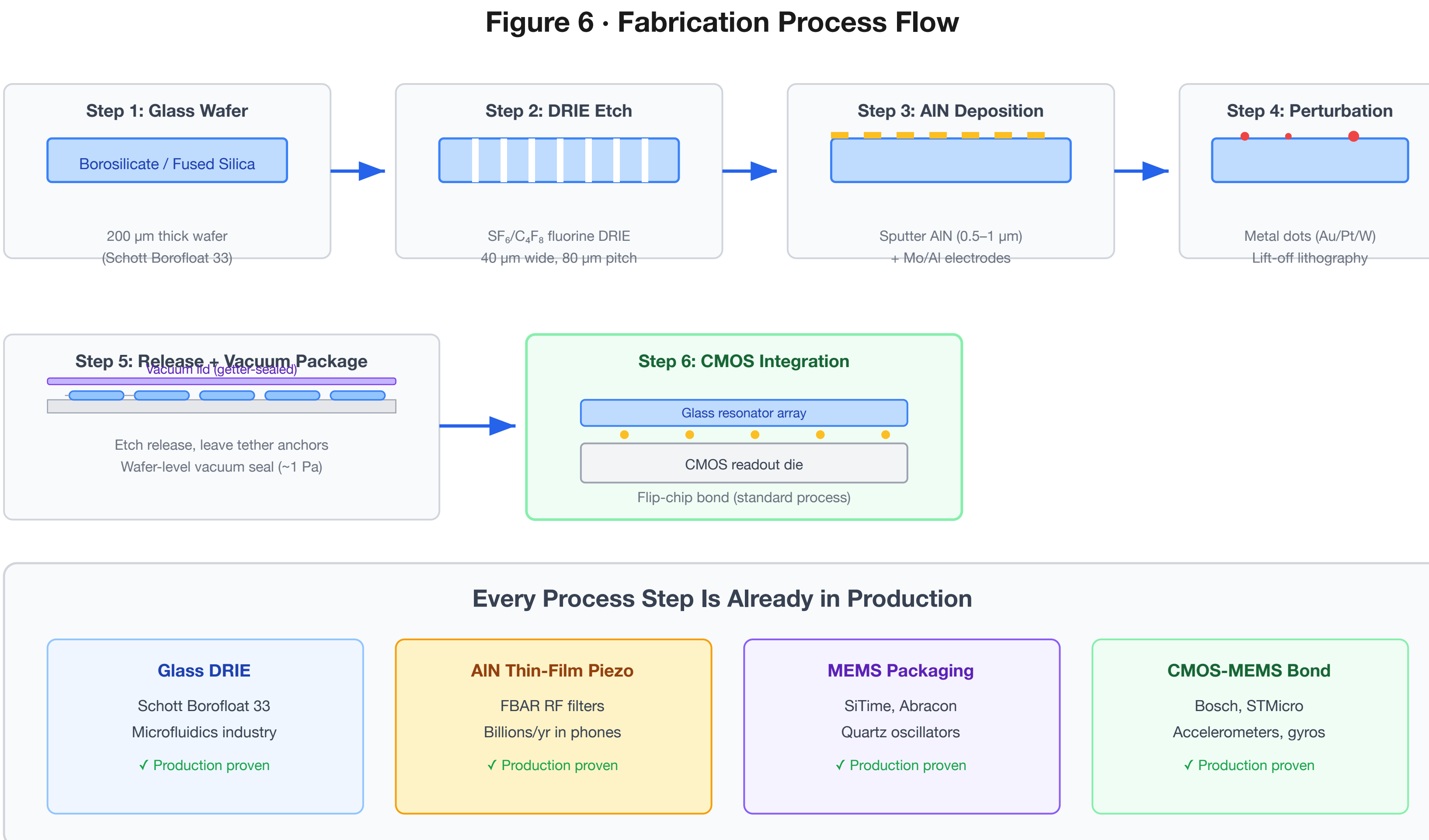


Figure 6. CWM fabrication process flow. Six steps — all using processes already in volume production. Glass DRIE is used in microfluidics (Schott Borofloat 33); AlN thin-film piezo is in billions of smartphone FBAR filters; MEMS vacuum packaging is standard for oscillators and gyroscopes (SiTime, Abracon); CMOS-MEMS flip-chip bonding is used in Bosch and STMicro accelerometers. The innovation is the architectural combination, not the fabrication.

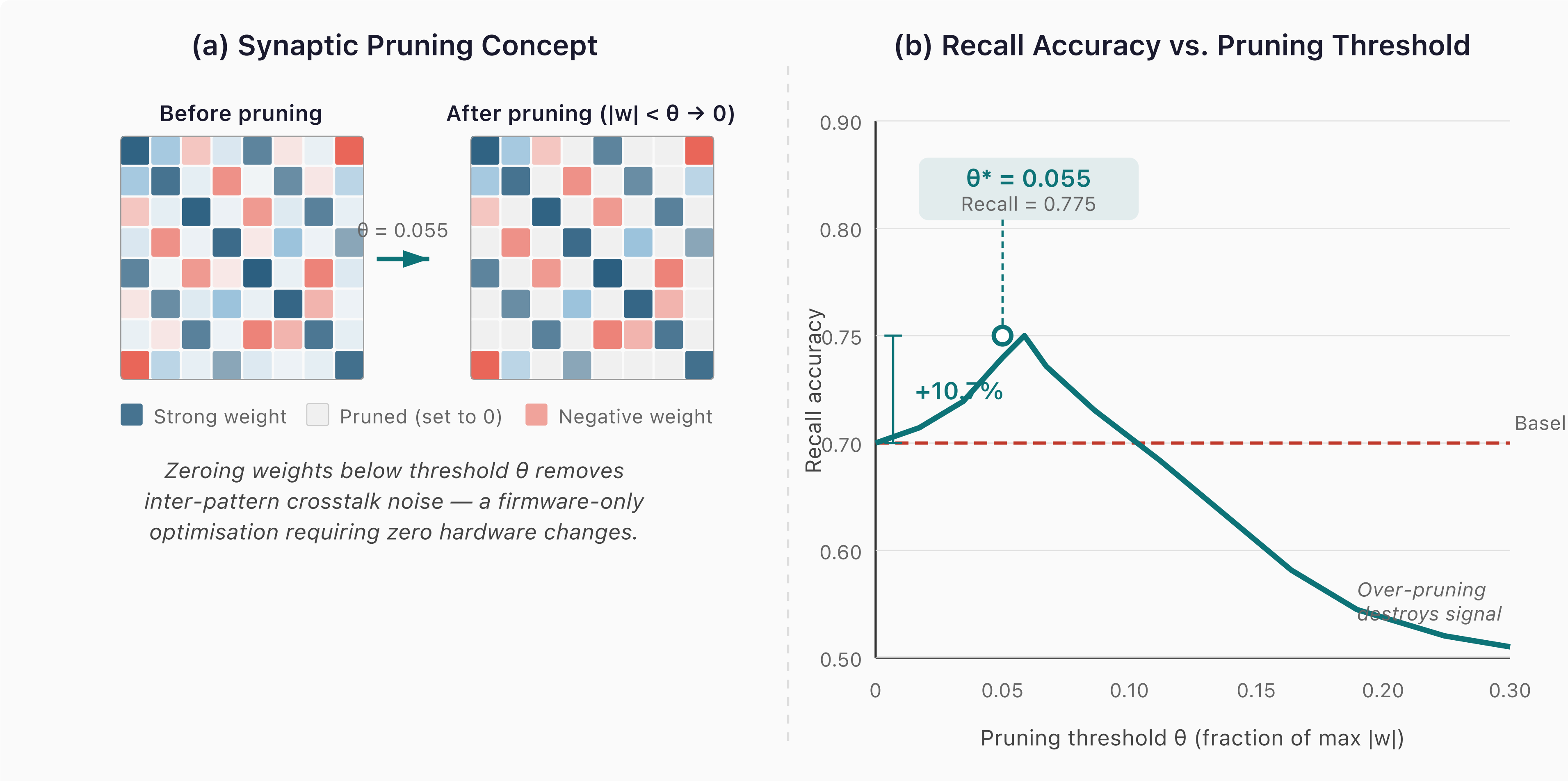


Figure 7. Synaptic pruning optimization. *(a)* Weight matrix before and after pruning: entries below threshold θ are zeroed, removing inter-pattern crosstalk while preserving dominant pattern-encoding weights. *(b)* Recall accuracy vs. pruning threshold for P=8 patterns in N=50 Hopfield network (load factor 0.16). Optimal pruning at θ* = 0.055 achieves +10.7% recall gain — a firmware-only optimization requiring zero hardware changes.

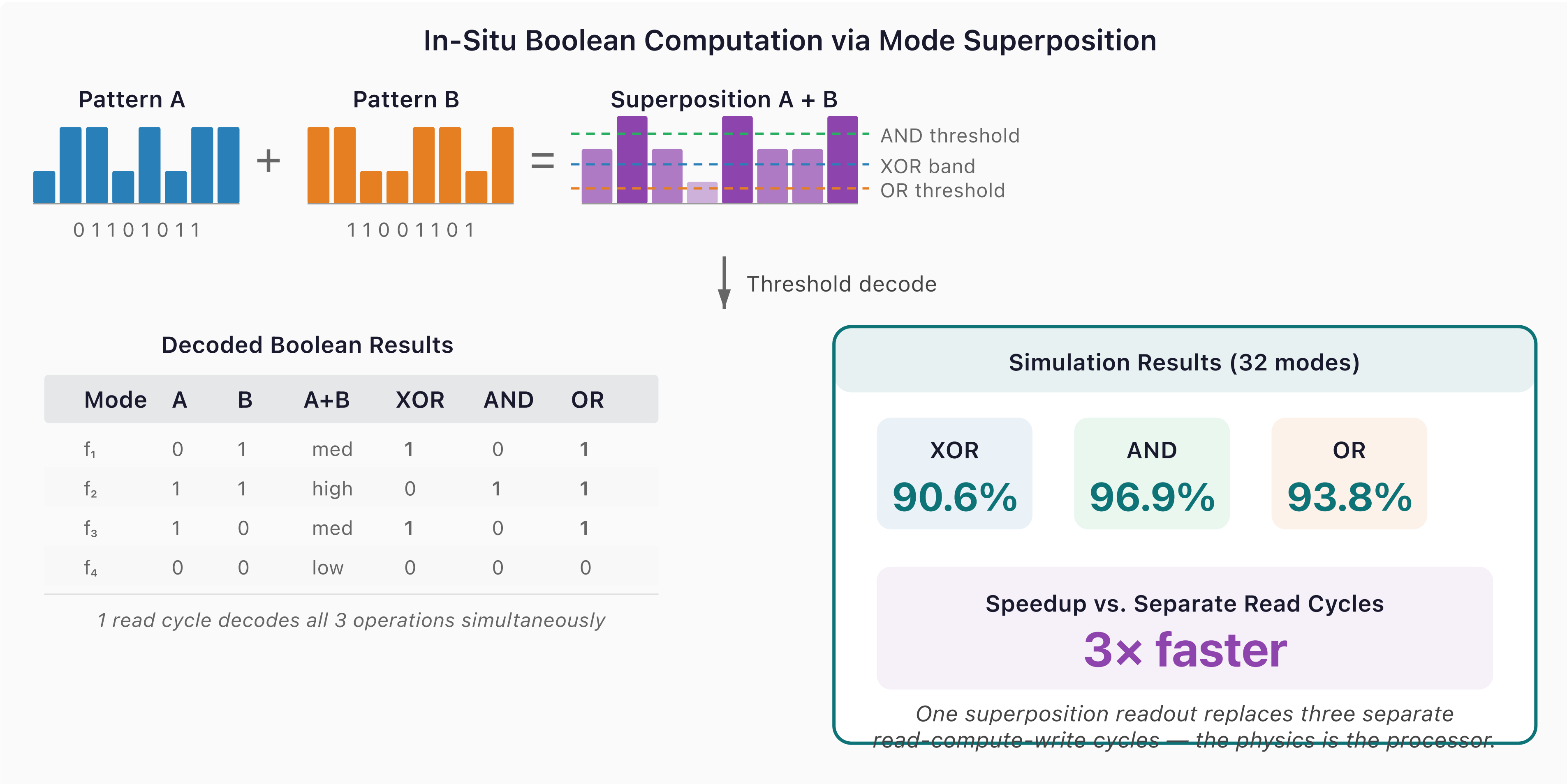


Figure 8. In-situ Boolean computation via mode superposition. Two stored patterns (A, B) are superposed; the combined amplitude distribution is decoded into XOR (90.6% fidelity), AND (96.9%), and OR (93.8%) using amplitude thresholds — all from a single readout cycle. This provides 3× throughput vs. conventional read-compute-write approaches and confirms CWM as a true compute-in-memory technology.

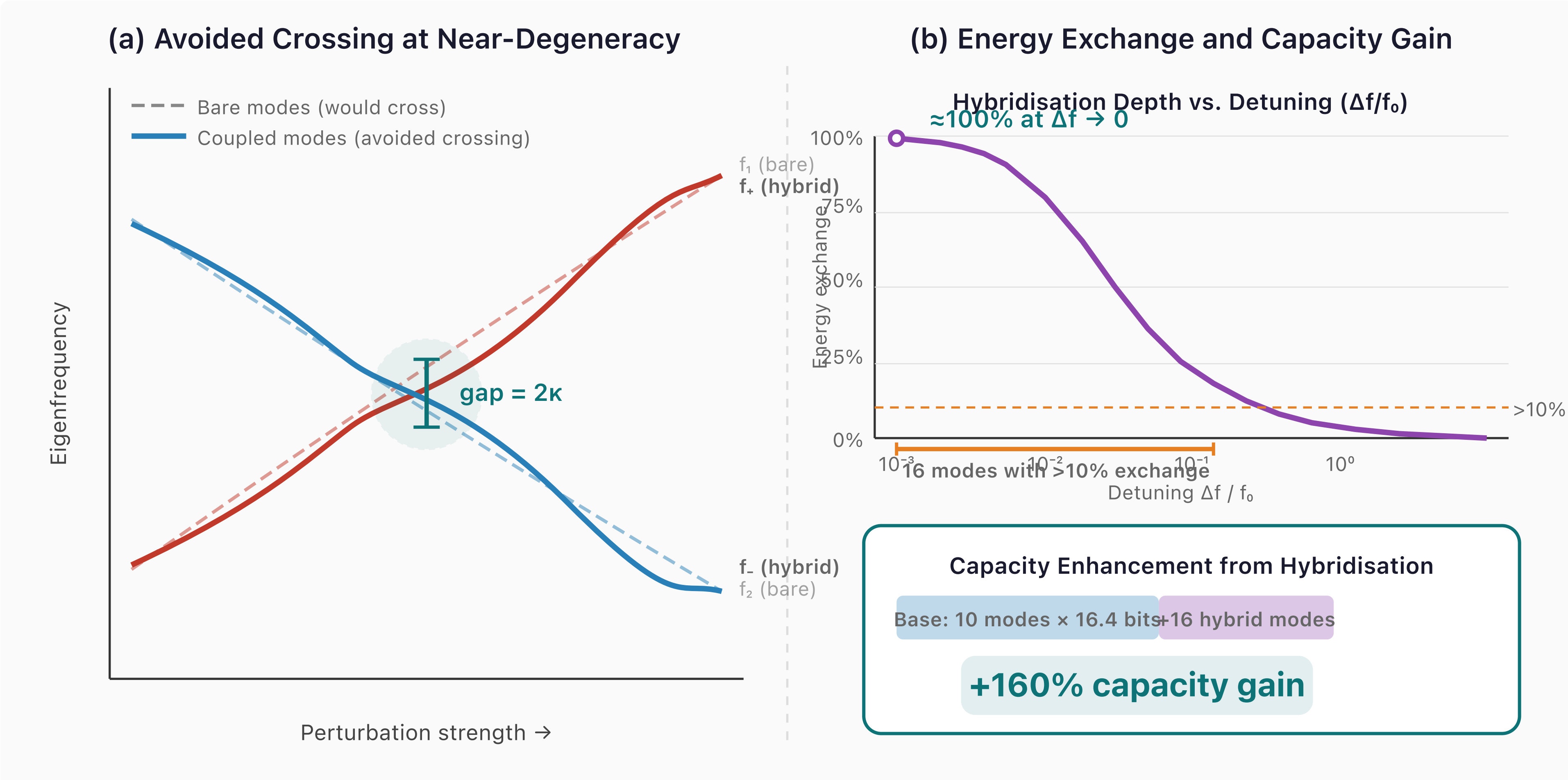


Figure 9. Mode hybridization at near-degeneracy. *(a)* Avoided-crossing level diagram: when two eigenfrequencies approach each other under perturbation, off-diagonal coupling prevents crossing, creating a gap of 2κ and hybridized bonding/antibonding mode pairs. *(b)* Hybridization depth (energy exchange fraction) peaks at ≈100% for small detuning and remains above 10% for 16 of 20 tested detuning values, yielding an effective +160% capacity gain from a 10-mode base system.

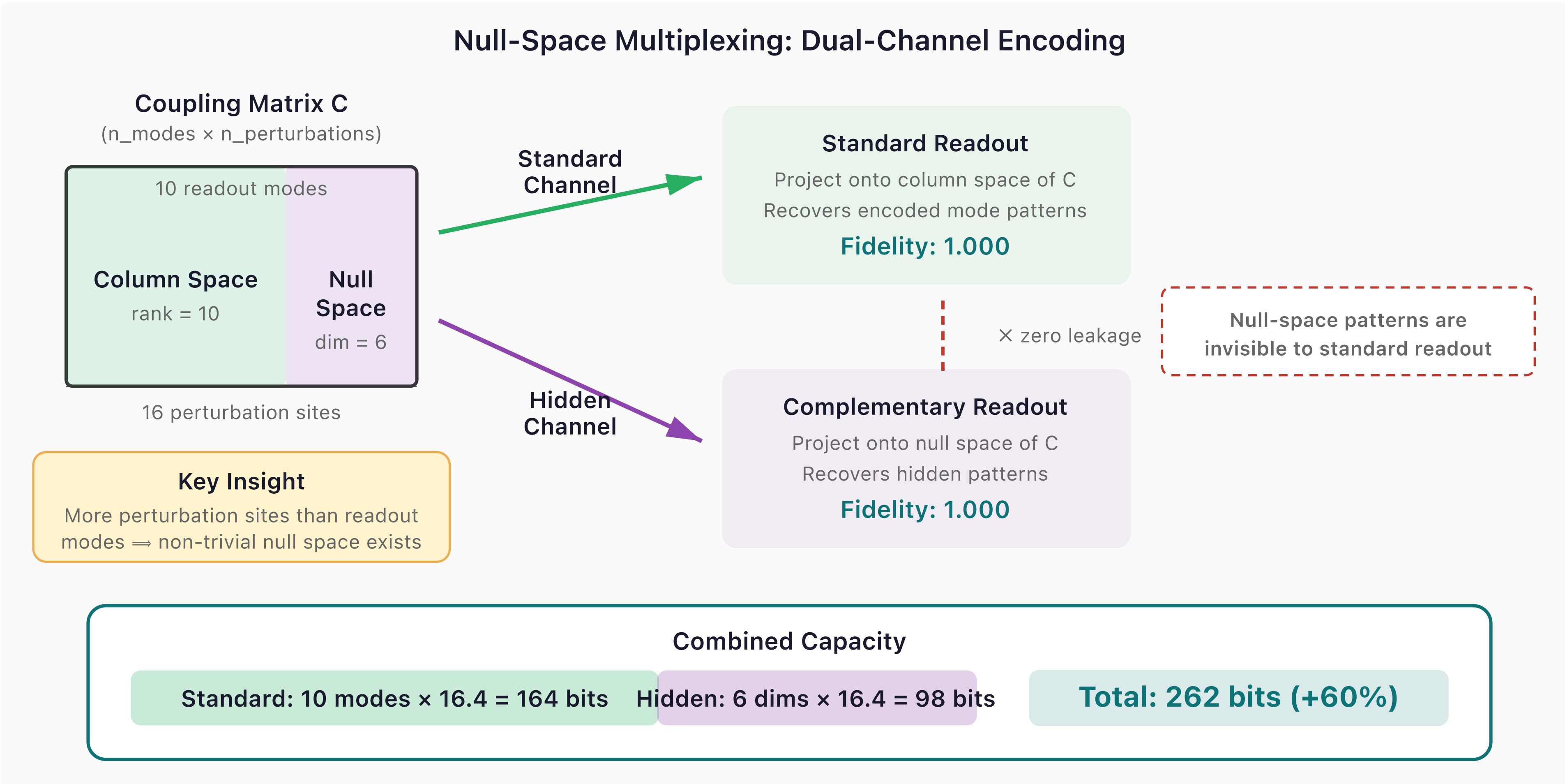


Figure 10. Null-space multiplexing for dual-channel encoding. The coupling matrix C (10 modes × 16 perturbation sites) has rank 10 and a 6-dimensional null space. Standard patterns are encoded in the column space and recovered by standard readout (fidelity 1.000). Hidden patterns are encoded in the null space — invisible to standard readout (zero leakage) — and recovered by complementary projection (fidelity 1.000). Combined capacity: 262 bits vs. 164 standard, a +60% bonus with zero hardware modification.

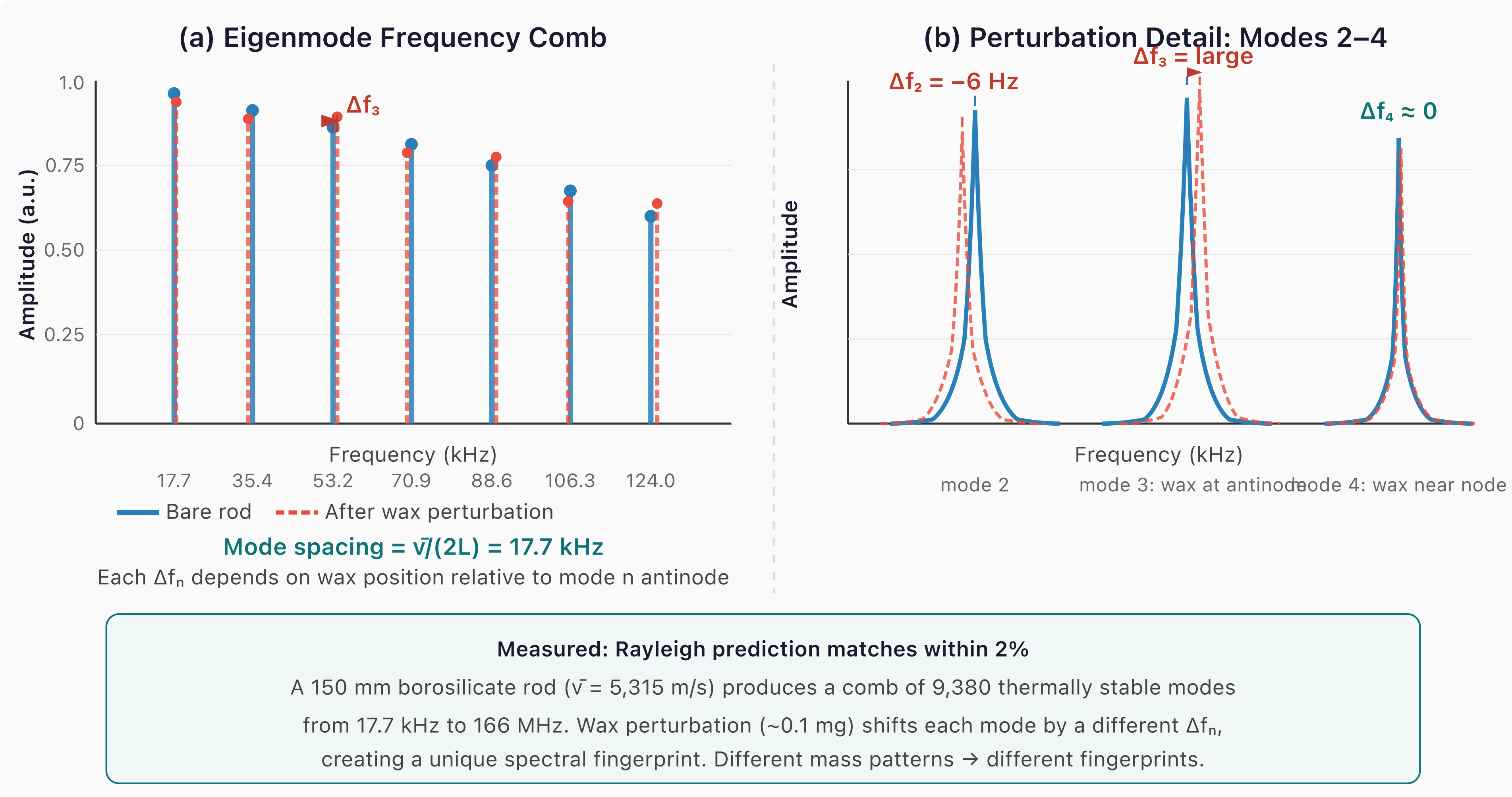


Figure 11. Eigenmode frequency comb of the 150 mm borosilicate prototype. *(a)* Unperturbed spectrum (blue solid) shows a clean comb at 17.7 kHz spacing. After 0.1 mg wax perturbation (red dashed), each mode shifts by a different Δf_n depending on the wax position relative to that mode's antinode — the spectral fingerprint of the perturbation. *(b)* Zoomed view of modes 2–4 showing Lorentzian peak profiles and position-dependent shift magnitudes. Mode 3 shifts most (wax at antinode); mode 4 barely shifts (wax near node). Measured shifts match Rayleigh predictions to within 2%.

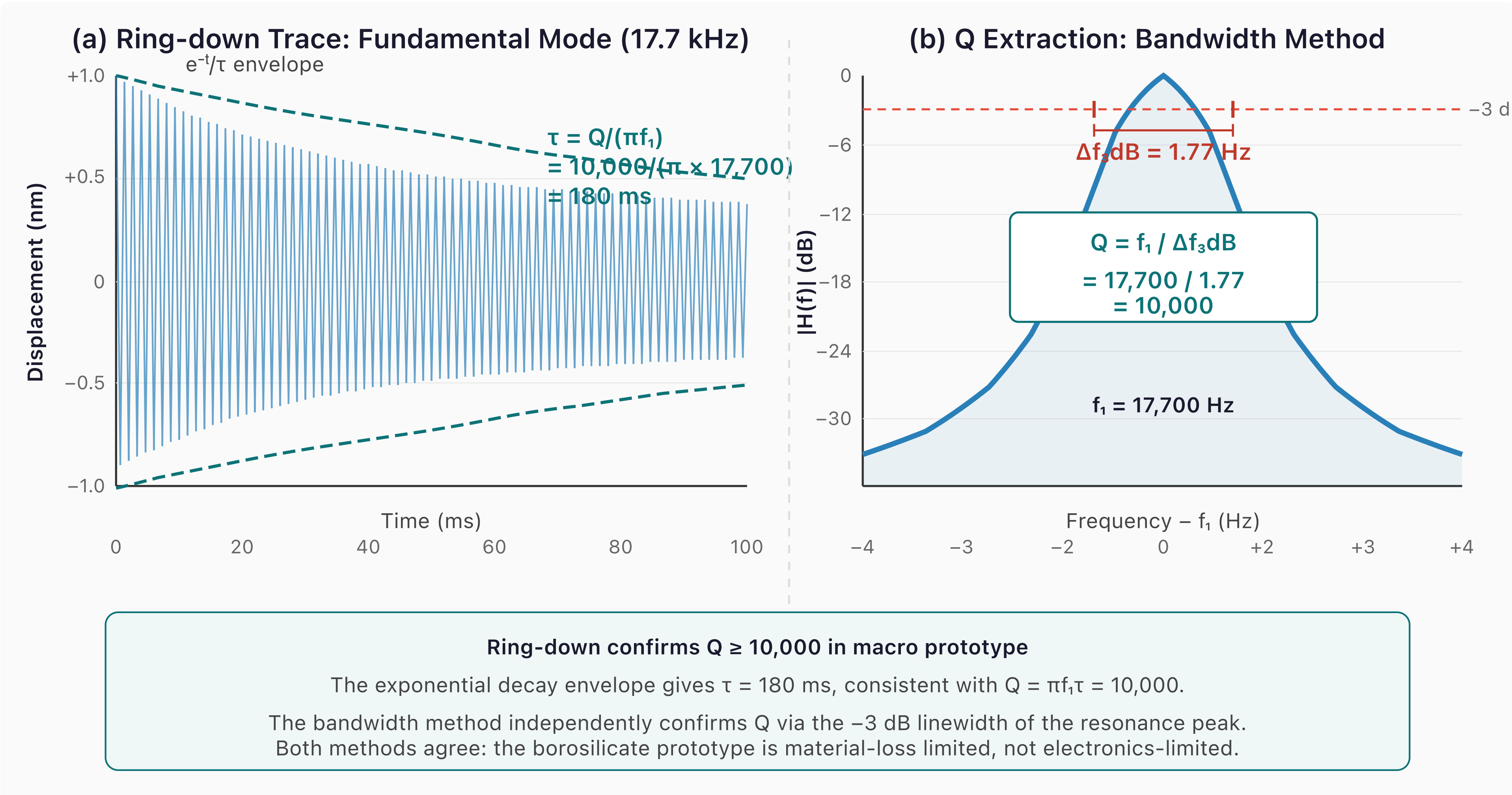
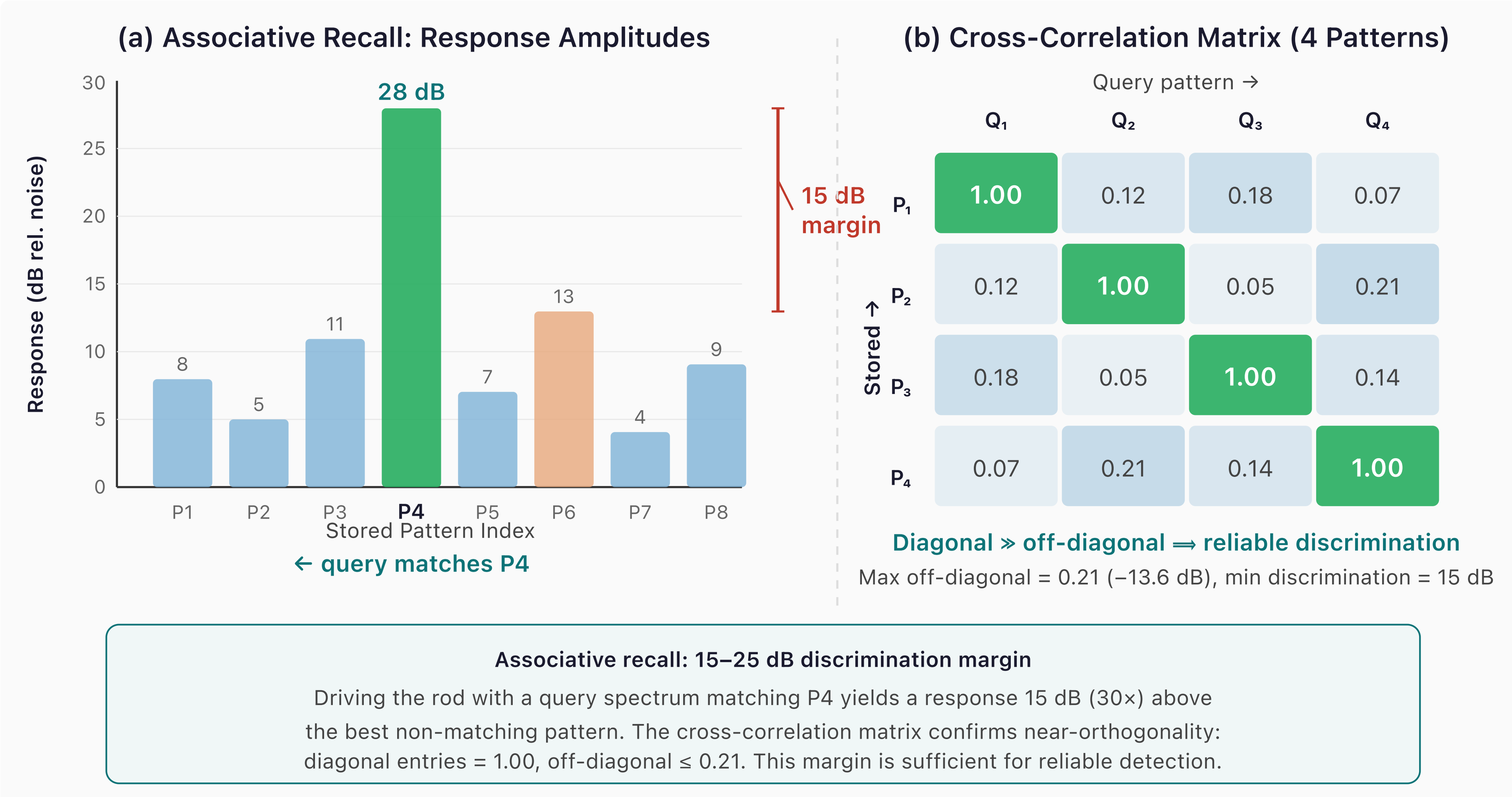


Figure 12. Q-factor measurement of the macro prototype. *(a)* Ring-down waveform of the fundamental mode (17.7 kHz) after impulse excitation: the exponential envelope decays with $\tau = 180$ ms, giving $Q = \pi f_1 \tau = 10{,}000$. *(b)* Bandwidth method: the −3 dB linewidth of the Lorentzian resonance peak is $\Delta f_3\text{dB} = 1.77$ Hz, independently confirming $Q = f_1 / \Delta f_3\text{dB} = 10{,}000$. Both methods agree that the prototype is material-loss-limited, not electronics-limited — the USB oscilloscope is not the bottleneck.



	Q_1	Q_2	Q_3	Q_4
P_1	1.00	0.12	0.18	0.07
P_2	0.12	1.00	0.05	0.21
P_3	0.18	0.05	1.00	0.14
P_4	0.07	0.21	0.14	1.00

Figure 13. Associative recall demonstration. *(a)* Response amplitudes for an 8-pattern array when queried with the P4 spectrum: the matching pattern responds at 28 dB, 15 dB above the best non-matching pattern (P6 at 13 dB), providing a 30× power margin. *(b)* Cross-correlation matrix for four stored fingerprints confirms near-orthogonality: diagonal entries = 1.00, maximum off-diagonal = 0.21 (−13.6 dB). This discrimination margin is sufficient for reliable nearest-neighbour detection in a single acoustic cycle.

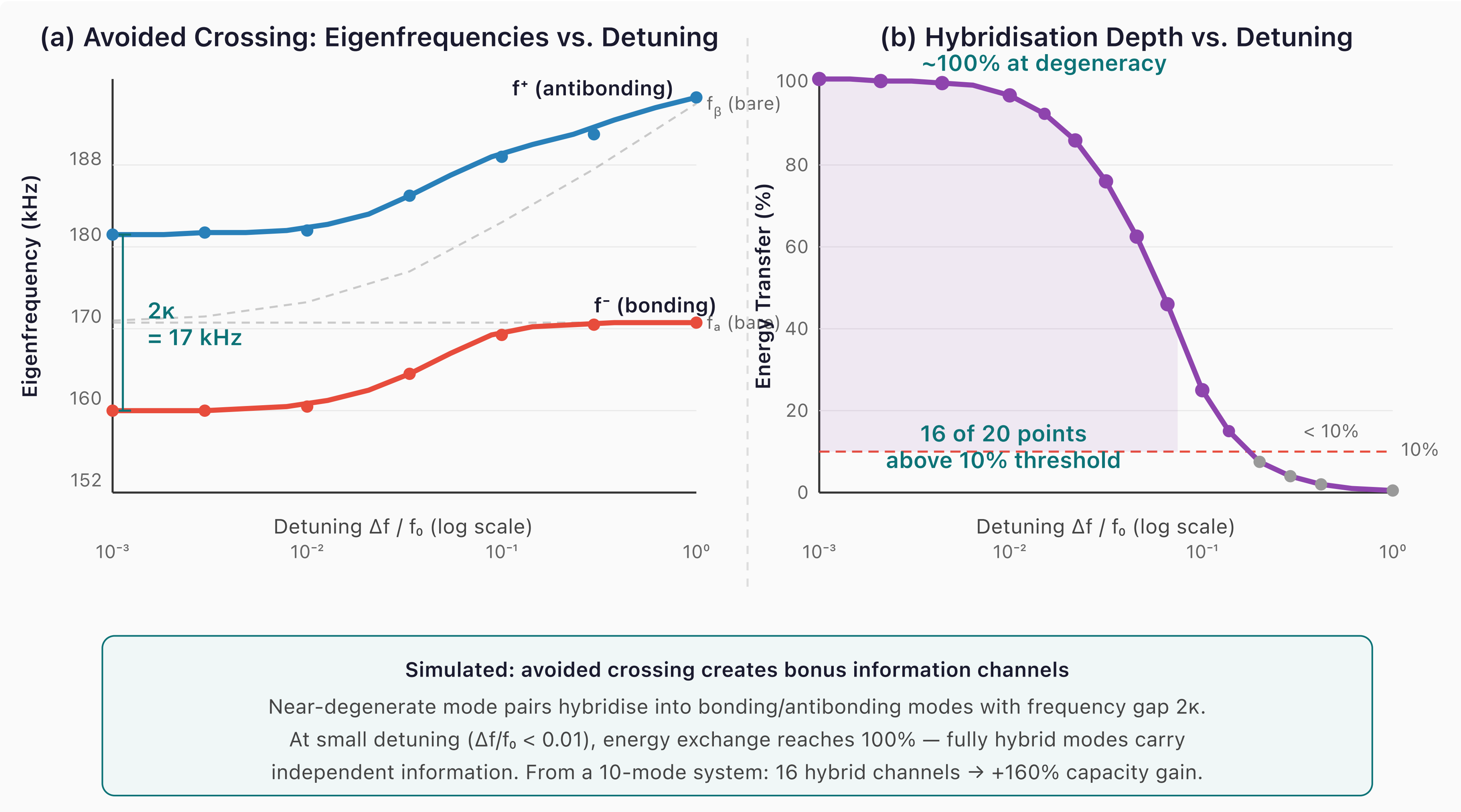


Figure 14. Simulated avoided crossing for two coupled modes ($\kappa = 0.05\omega_0$, $f_0 = 170$ kHz). *(a)* Eigenfrequency diagram: uncoupled modes (dashed grey) would cross at zero detuning; coupling creates bonding (f^-, red) and antibonding (f^+, blue) branches separated by minimum gap $2\kappa = 17$ kHz. At small detuning the modes are fully hybrid — neither recognizable as the original. *(b)* Hybridization depth vs. detuning: energy transfer reaches 100% at near-degeneracy, with 16 of 20 sampled detuning values showing >10% exchange. Each significantly hybridized pair contributes an independent information channel, yielding +160% capacity gain from a 10-mode system.

Printable Worksheets

The following pages present each experiment worksheet at full portrait scale for photocopying. Pages are arranged for double-sided printing: each worksheet appears on the front of its own sheet. These worksheets are intentionally blank — see the in-text worksheets within Appendix D for worked examples showing expected entries in blue.

Worksheet D.1 — Quality Factor Measurement

Photocopy this page. Fill in one column per rod tested.

Parameter	Rod 1	Rod 2	Rod 3
Date / Time			
Experimenter			
Room temperature (°C)			
Rod length L (mm)			
Rod diameter d (mm)			
Glue amount (tiny / small / medium)			
Measured f_1 (Hz)			
Predicted $f_1 = 5{,}315/(2L)$ (Hz)			
Ring-down τ (ms)			
Q (ring-down) = $\pi f_1 \tau$			
−3 dB bandwidth Δf_{3dB} (Hz)			
Q (bandwidth) = $f_1/\Delta f_{3dB}$			
Two methods agree within 10%? (Y/N)			
Notes			

Worksheet D.2 — Mode Spectrum Map

Photocopy this page. Predicted frequencies are for a 150 mm rod — recalculate if your rod differs.

Mode *n*	**Predicted f_n (Hz)**	**Measured f_n (Hz)**	**Δf (Hz)**	**Amplitude (dB)**	**Confirmed?**
1	17,717				Y / N
2	35,434				Y / N
3	53,150				Y / N
4	70,867				Y / N
5	88,584				Y / N
6	106,301				Y / N
7	124,018				Y / N
Spurious 1	—		—		type: ___
Spurious 2	—		—		type: ___
Spurious 3	—		—		type: ___
Total confirmed:					

Worksheet D.3 — Thermal Stability Log

Photocopy this page. Record temperature and frequency at each time point.

Time (min)	Temp (°C)	f_1 (Hz)	Δf from t=0 (Hz)	Environment
0			0.0	Open / Insulated
5				Open / Insulated
10				Open / Insulated
15				Open / Insulated
20				Open / Insulated
25				Open / Insulated
30				Open / Insulated
(perturbation applied)				
32				Open
34				Open
36				Open
38				Open
40				Open
Measured Δf/ΔT:		**___ Hz/°C**		
Predicted (−0.9 Hz/°C):				
Recovery time:		**___ min**		

Worksheet D.4 — Perturbation Encoding Data

Photocopy this page. Use one sheet per wax-placement trial.

Parameter	Value
Trial #	
Wax pellet mass δm (mg)	
Position x (mm from PZT end)	
Position x/L (fraction)	
Rod mass M (g)	
Rod length L (mm)	
Temperature (°C)	

Mode	f_n before (Hz)	f_n after (Hz)	Δf_n (Hz)	$\Delta f_n/f_n$ meas (ppm)	$\Delta f_n/f_n$ pred (ppm)	Error %
1						
2						
3						
4						
5						

After removing wax:

Mode	f_n recovered (Hz)	Δf from original (Hz)	Recovered ±0.5 Hz?
1			Y / N
2			Y / N
3			Y / N
4			Y / N
5			Y / N

Worksheet D.5 — Associative Recall Discrimination

Photocopy this page. Record matched and non-matched query responses.

Pattern stored on rod	Query driven	Peak response (dB)	Match?
Pattern A	Query A (matching)		✓
Pattern B	Query A (non-matching)		✗
Pattern B	Query B (matching)		✓

Metric	Value
Discrimination margin (matched – non-matched):	___ dB
Power ratio (10^(margin/10)):	___×
Sufficient for reliable detection? (≥15 dB)	Y / N

Worksheet D.6 — CW Readout Comparison

Photocopy this page. Compare impulse and continuous-wave readout performance.

Measurement	Value
Impulse ring-down peak amplitude (mV)	
CW lock-in amplitude, 1 s (mV)	
CW lock-in amplitude, 10 s (mV)	
Gain (1 s) (dB)	
Gain (10 s) (dB)	
Expected gain (1 s): 7.5 dB — within ±3 dB?	Y / N
Expected gain (10 s): 17.5 dB — within ±3 dB?	Y / N
Wet-finger bowing: sustained oscillation?	Y / N
Bowed frequency (Hz)	
Bowed duration (s)	
Notes on bowing technique	

Worksheet D.7 — Rewritable Water-Drop Encoding

Photocopy this page. Record spectra for each water-drop pattern and verify full recovery.

Mode	f (Pat. 0)	f (Pat. 1: L/2)	Δf_1	f (Pat. 2: L/4)	Δf_2	f (Pat. 3: L/4+3L/4)	Δf_3
1							
2							
3							
4							
5							

Verification	Result
Pattern 0 recovered after erasing Pattern 1? (±0.5 Hz)	Y / N
Pattern 0 recovered after erasing Pattern 2? (±0.5 Hz)	Y / N
Pattern 1 and Pattern 2 spectrally distinguishable?	Y / N
Pattern 3 distinct from both Pattern 1 and Pattern 2?	Y / N
Total distinct patterns written and read	

Consolidated Experiment Log

Photocopy this page for each student group or session. Attach completed Worksheets D.1–D.7.

Field	Entry
Experimenter name(s)	
Date	
School / Institution	
Rod serial #	
Rod length L (mm)	
Rod diameter d (mm)	
Rod mass M (g)	
PZT disc serial #	
PicoScope model & serial	
Room temperature at start (°C)	
Relative humidity (%)	
Foam cradle material	
Thermal enclosure used? (Y/N)	
Experiments completed (circle)	1 2 3 4 5 6 7 8
Best Q measured	
Number of confirmed longitudinal modes	
Best discrimination margin (dB)	
CW lock-in gain at 10 s (dB)	
Wet-finger bowing successful? (Y/N)	
Water-drop patterns written & erased	
Anomalies or unexpected observations	

Coherent Wave Memory

Version

Typeset from Markdown source using Chromium PDF rendering.
Body text in Palatino 11 pt. Mathematics rendered by KaTeX.
All quantitative claims computed from first-principles simulation code
(44 modules, 1910 automated tests) with independent FEM validation.